A Morse-Theoretic Construction of the Atiyah–Hirzebruch Spectral Sequence

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Contents

Notation & Conventions

This page collects all notational conventions used in this thesis. It is to be **removed before submission**.

1. Manifolds, Maps, and the Differential

Symbol	Meaning & Convention
M, N	smooth manifolds (always assumed without boundary unless stated)
T_pM, T_p^*M	tangent / cotangent space at $p \in M$
TM, T^*M	tangent / cotangent bundle
$\mathrm{d}f$	exterior differential of $f\colon M \to \mathbb{R}$; a 1-form on M
$\mathrm{d}_pf = (\mathrm{d}f)_p$	differential of f *at the point* p; a linear map $T_pM \to \mathbb{R}$. **Rule:** write d_pf (macro `\dat{p}f`) when evaluating at a specific point. Write $\mathrm{d}f$ for the full 1-form / bundle map.
d_pF	for $F\colon M \to N$: the linear map $T_pM \to T_{F(p)}N$. Also written $(\mathrm{d}F)_p$; never write $\frac{\mathrm{d}}{\mathrm{d}x}F$ for a map between manifolds.
$\frac{\mathrm{d}}{\mathrm{d}t}$	total derivative w.r.t. a *single* real variable t (roman d, macro `\diff`). Use for curves, flows evaluated on a trajectory.
$\frac{\partial}{\partial t}$	partial derivative; use only when *at least two independent* variables/parameters are present. **Never** write $\partial/\partial t$ for the time-derivative along a flow line.
$\nabla_v f$	covariant derivative of f in direction v (Levi-Civita or chosen connection)
$\mathrm{grad}_g f$	gradient of f with respect to Riemannian metric g; characterised by $g(\mathrm{grad}_g f, -) = \mathrm{d}f$

2. Flows and Dynamical Systems

Symbol	Meaning & Convention
$\varphi_t(x)$	flow of a vector field at time t through initial point x. **Convention:** always write φ_t, not $\varphi(t,\cdot)$; think of $\varphi_t \in \mathrm{Diff}(M)$ as a one-parameter family of diffeomorphisms.
$\varphi_t = \mathrm{e}^{tX}$	flow of the vector field X; $\varphi_0 = \mathrm{id}$.
$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(x)$	time-derivative of the flow line $t \mapsto \varphi_t(x)$; equals $X(\varphi_t(x))$. Use roman d, never ∂.
$W^{\mathrm{u}}(p)$, $W^{\mathrm{s}}(p)$	unstable / stable manifold of a critical (rest) point p (macros `\Wu{p}`, `\Ws{p}`)

Symbol	Meaning & Convention
$\mathcal{M}(p,q)$	moduli space of (unparametrised) flow lines from p to q (macro `\Mflow{p}{q}`)

3. Restrictions and Evaluations

Core rule: distinguish carefully between a *fiber*, a *restriction to a subspace*, and a *value/evaluation.*

Symbol	Meaning & Convention
E_p	**fiber** of a vector bundle $E \to X$ over $p \in X$. **Preferred over** $E\|_{\{p\}}$ for fibers; use macro `E_p`.
$E\|_A$	**restriction** of the bundle E to the subspace $A \subseteq X$; this is itself a bundle over A. Macro: `\restr{E}{A}`.
$\xi\|_A$	section $\xi \in \Gamma(E)$ restricted to A; yields $\xi\|_A \in \Gamma(E\|_A)$.
$\xi(p)$ or ξ_p	**value** of the section ξ at the point p; an element of E_p. **Avoid** $\xi\|_p$ for this; use $\xi(p)$.
$\omega_p = \omega\|_p$	value of a differential k-form ω at p; an element of $\Lambda^k T_p^* M$. Acceptable to write $\omega\|_p$ only when the point-evaluation is being *emphasised* over the fiber.
Summary	E_p = fiber (bundle); $E\|_A$ = restricted bundle; $\xi(p)$ = value of section; ω_p = value of form.

5. Chain Complexes and Homology Theories

Three homology theories appear simultaneously; they are always decorated.

Theory	Chain complex	Boundary / Homology
Singular	$({}^{\mathcal{S}}C_*(X), {}^{\mathcal{S}}\partial)$	boundary ${}^{\mathcal{S}}\partial$; homology $H^{\mathrm{S}}H_*(X)$. Macros: `\ChS`, `\bS`, `\HS`.
CW	$({}^{\mathcal{CW}}C_*(X), {}^{\mathcal{CW}}\partial)$	boundary ${}^{\mathcal{CW}}\partial$; homology $H^{\mathrm{CW}}H_*(X)$. Macros: `\ChCW`, `\bCW`, `\HCW`.
Morse	$({}^{\mathcal{M}}C_*(f), {}^{\mathcal{M}}\partial)$	boundary ${}^{\mathcal{M}}\partial$; homology ${}^{\mathcal{M}}H_*(f)$. Macros: `\ChM`, `\bM`, `\HM`.

Connecting homomorphisms in long exact sequences: : ${}^{\mathcal{M}}\delta, {}^{\mathcal{CW}}\delta, {}^{\mathcal{S}}\delta$ with the macros: `\connM`, `\connCW`, `\connS`.
Co-homology: : For co-homology we use upper indices on the co-boundary maps: ${}^{\mathcal{M}}\mathrm{d}, {}^{\mathcal{CW}}\mathrm{d}, {}^{\mathcal{S}}\mathrm{d}$ with the macros: `\cbM`, `\cbCW`, `\cbS`. And for the co-homology we use the homlogie macros with upper indices.

Theory	Chain complex	Boundary / Homology
Filtration: $F^{p\mathcal{S}}C_*(X)$ for the filtration of the singular complex; boundary is still ∂.		

6. Vector Bundles and K-Theory

Symbol	Meaning
$\mathrm{Vect}(X)$	set of isomorphism classes of complex vector bundles over X (macro `\Kvect{X}`)
$[E]$	stable isomorphism class / element in $\mathrm{K}(X)$
$\mathrm{K}(X) = \mathrm{KU}(X)$	complex K-theory (default); macro `\K`, `\KU`
$\mathrm{KO}(X)$	real K-theory; macro `\KO`
$\tilde{\mathrm{K}}(X)$	reduced complex K-theory
$E \oplus F,\ E \otimes F$	Whitney sum / tensor product of bundles
$E\vert_A$	restriction of bundle $E \to X$ to subspace A (same convention as §3)
E_p	fiber of E over p (same convention as §3)

7. Spectral Sequences

Symbol	Meaning
$E^r_{p,q}$	(p,q)-entry on the r-th page of a spectral sequence; macro `\Epq{r}{p,q}`
d^r	differential on the r-th page; $d^r\colon E^r_{p,q} \to E^r_{p-r,\,q+r-1}$ (cohomological convention); macro `\drpq{r}`
AHSS	Atiyah–Hirzebruch spectral sequence: $E^2_{p,q} \cong H^p(X; \mathrm{K}^q(\mathrm{pt})) \Rightarrow \mathrm{K}^{p+q}(X)$
$E^\infty_{p,q}$	abutment / E_∞-page

8. Frequently Used Macros (Quick Reference)

Macro	Output	Use
`\diff`	d	exterior deriv. / total deriv.
`\dat{p}`	d_p	deriv. at point p
`\restr{E}{A}`	$E\vert_A$	restriction
`\Wu{p}`	$W^{\mathrm{u}}(p)$	unstable manifold

Macro	**Output**	**Use**
`\Ws{p}`	$W^{\mathrm{s}}(p)$	stable manifold
`\Mflow{p}{q}`	$\mathcal{M}(p,q)$	moduli of flow lines
`\ChM`	${}^{\mathcal{M}}C_*$	Morse chain complex
`\bM`	${}^{\mathcal{M}}\partial$	Morse boundary
`\HM`	${}^{\mathcal{M}}H_*$	Morse homology
`\ChCW`	${}^{\mathcal{CW}}C_*$	CW chain complex
`\bCW`	${}^{\mathcal{CW}}\partial$	CW boundary
`\HCW`	$H^{\mathrm{CW}}H_*$	CW homology
`\conn`	δ	connecting homom. (δ)
`\K`	K	K-theory
`\KO`	KO	real K-theory
`\KU`	KU	complex K-theory
`\Epq{r}{p,q}`	$E^r_{p,q}$	SS entry
`\drpq{r}`	d^r	SS differential
`\Crit`	Crit	critical point set
`\ind`	ind	Morse index
`\grad`	grad	gradient
`\euler`	e	Euler number (roman e)
`\ii`	i	imaginary unit (roman i)

1 Introduction

2 Vector Bundles and Riemannian Geometry

sec:background-geometry

Due to differential geometry and the theory of vector bundles being areas of mathematics used by pure mathematicians interested in analytical aspects like curvature, by topologists interested in invariants like Euler characteristics, physicists using the spaces as playgrounds for general relativity theories, and many others, there are numerous competing notational conventions. This thesis lies at the tripoint of differential geometry, dynamical systems, and algebraic topology. Hence, there is no a priori preferred notational convention. In this chapter, we establish the necessary background in the theory of vector bundles and Riemannian manifolds whilst not proving all the basics; we establish the notation used throughout this thesis. We establish manifolds and vector bundles in parallel, as they are interdependent on one another.

2.1 Differentiable Structures on Topological Manifolds

We begin by defining a topological manifold. Afterwards, we equip that space with with increasingly more structure and hence reduce the generality until we arrive at the playground of the thesis:

def:vb-topologicalManifold

Definition 2.1.1 (Topological Manifold)**.** A second countable Hausdorff space M is called a **topological manifold** of dimension $m \in \mathbb{N}$, if it is locally homeomorphic to $\mathbb{R}^m$. To be precise, for all $p \in M$ there exists an open neighbourhood $U \subseteq M$ of p, an open set $V \subseteq \mathbb{R}^m$ and a map $\varphi : W \to V$ that is a homeomorphism. We call the map $\varphi : U \to V$ a **chart around** p **on** M, and φ^{-1} a **local coordinate system around** p **on** M.

def:vb-differentiableManifold

Definition 2.1.2 (Differentiable Manifold)**.** Let M be a topological manifold of dimension m.

1. A **differentiable atlas of class** $r \in \mathbb{N} \cup \{\infty\}$ is a family of charts $\mathfrak{A} = (\varphi_i : U_i \to V_i)_{i \in I}$ such that
   a) $\bigcup_{i \in I} U_i = M$, meaning that (U_i) is an open covering of M.
   b) For every pair $(i, j) \in I^2$, the **transition function**:

$$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\to \varphi_i(U_i \cap U_j) \\ x &\mapsto \left(\varphi_i \circ \varphi_j^{-1}\right)(x) \end{aligned}$$

   is differentiable of class r.

   We call such an atlas a C^r-atlas.

2. Two C^r-atlases $\mathfrak{A}$ and $\mathfrak{B}$ are called **equivalent** if the family $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$ is a C^r-atlas.

A **differentiable structure of class** r on M is an equivalence class c of C^r-atlases. For $r = \infty$, we call the pair (M, c) a smooth manifold.

Corollary 2.1.3. Every transition function φ_{ij}, $i,j \in I^2$ of a differentiable atlas $\mathfrak{A} = (\varphi_i)_{i\in I}$ is not merely a homeomorphism but also a diffeomorphism. This follows from the fact that $\varphi_{ij}^{-1} = \varphi_{ji}$ which provides smooth inverses.

def:vb-smooth Functions

Definition 2.1.4 (Smooth Functions)**.** Let (M,c) be a differentiable manifold of class r and $U \subseteq M$ open. We call a continuous function

$$f : U \to \mathbb{R}$$

differentiable of class r, if for any chart (and hence for all) $(\varphi_i)_{i\in I} = \mathfrak{A} \in c$ the compositions $f \circ \varphi_i^{-1}$ are differentiable of class r. For $r = \infty$ we define:

$$\mathcal{E}(U) = \{f : U \to \mathbb{R} \text{ continuous} \mid f \text{ is differentiable of class } \infty\}.$$

Corollary 2.1.5. Let (M,c) be a smooth manifold of dimension m and $U \subseteq M$ be an open subset. With pointwise defined operations, the set $(\mathcal{E}(U),+,\cdot,\circ)$ becomes an $\mathbb{R}$−algebra. Furthermore, $\mathcal{E}$ becomes a sheaf of $\mathbb{R}$-algebras.

Proof. There is not really a need for a proof. However, it might help to work through the definition of a sheaf. First, $\mathcal{E}$ is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned} \operatorname{res}_V^U : \mathcal{E}(U) &\to \mathcal{E}(V) \\ f &\mapsto f|_V \,. \end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property. □

Definition 2.1.6. If $p \in M$ is fix, $f \in \mathcal{E}(U)$ and $g \in \mathcal{E}(U')$ such that $p \in U \cap U'$ we say that f and g have the same **germ in** p, if there is another open neighborhood $W \subseteq U \cap U'$ of p such that $f|_W = g|_W$. This defines an equivalence relation $\sim_p$. An equivalence class s of local functions around p is called a **germ in** p. We write $s = f_p$, if $s = [f]$ with $f \in \mathcal{E}(U)$. We write

$$\mathcal{E}_p(M) = \left(\sum_{U \text{open}, p\in U} \mathcal{E}(U) \right) \Big/ \sim_p$$

for the set of germs, and call it the **stalk in** p. Where $\sum$ denotes the co-product (disjoint union) in **Top**.

Corollary 2.1.7. For a smooth manifold (M,c) the set $\mathcal{E}_p(M)$ inherits an $\mathbb{R}$−algebra structure from the $\mathcal{E}(U)$. Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned} \operatorname{eval}_p : \mathcal{E}_p(M) &\to \mathbb{R} \\ f_p &\mapsto f(p) =: f_p(p) \end{aligned}$$

The stalks are also local rings with maximal ideal $\mathfrak{m}_p = \ker(\operatorname{eval}_p)$. Hence, the pair $(M,\mathcal{E})$ gives us a locally ringed space.

Proof. We only need to verify the local ring structure of the stalks. An element $f_p \in \mathcal{E}_p(M)$ is invertible if and only if $f(p) \neq 0$, equivalently if $f_p \notin \ker(\mathrm{eval}_p)$. Thus $\mathcal{E}_p(M)\big/\ker(\mathrm{eval}_p)$ is a field, which proves that $\ker(\mathrm{eval}_p)$ is maximal. □

Definition 2.1.8. Let (M,c) be a smooth manifold of dimension m and $p \in M$. We call an $\mathbb{R}$-linear map $\delta : \mathcal{E}_p(M) \to \mathbb{R}$ a **derivation**, if it satisfies the Leibniz rule:

$$\delta(f_p \cdot g_p) = \delta(f_p)g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f_p, g_p \in \mathcal{E}_p(M)\,.$$

We call $\mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M),\mathbb{R})$ the set of derivations and give it an $\mathbb{R}$-vector space structure by pointwise operations. We define the **tangent space of** M **at** p to be the vector space

$$T_pM \coloneqq \mathrm{Der}_{\mathbb{R}}(\mathcal{E}_p(M),\mathbb{R})\,.$$

Corollary 2.1.9. Let (M,c) be a smooth manifold of dimension m and $\varphi : U \to V$ be a chart around p with $x_0 = \varphi(p)$ (where $\varphi \in \mathfrak{A} \in c$). Then we define the derivation

$$\begin{aligned} \partial_i(p) : \mathcal{E}_p(M) &\to \mathbb{R} \\ f_p &\mapsto \partial_i(p)(f_p) \coloneqq \frac{\partial(f \circ \varphi^{-1})}{\partial x_i}\Big|_{x_0}\,. \end{aligned}$$

This is well-defined and hence gives a tangent vector. Moreover, the family

$$(\partial_1(p), \dots, \partial_m(p))$$

defines a basis of T_pM. Hence, the dimension of T_pM is m.

This is a good point to take a break from smooth manifolds and introduce the structure of vector bundles.

Analogously to the definition of a smooth manifold, we start by defining families of vector spaces and equip them with increasingly more structure:

`def:vb-vectorBundles`

Definition 2.1.10 (Vector Bundles)**.** A **complex vector bundle** of rank m is a surjection

$$\pi : E \to X$$

such that E and X are topological spaces and π is continuous. Furthermore π satisfies:

1. For all $x \in X$, $\pi^{-1}(x)$ carries the structure of an m-dimensional $\mathbb{C}$-vector space.
2. For all $x \in X$, there exist a neighbourhood $U \subseteq X$ and a homeomorphism

   $$\psi : E|_U \coloneqq \pi^{-1}(U) \to U \times \mathbb{C}^m\,,$$

   that is compatible in the following sense: Let π_1 denote the projection $U \times \mathbb{C}^m \to U$ onto the first factor. Then

   $$\pi_1 \circ \psi = \pi$$

   and

   $$\psi_x : E_x \coloneqq \pi^{-1}(x) \to \{x\} \times \mathbb{C}^m$$

   is a vector space isomorphism.

We call E the **total space**, X the **base space**, E_x the **fibre over** x and ψ a **local trivialization**.

Definition 2.1.11. Let $\pi : E \to X$ be a vector bundle. A continuous function $s : X \to E$ such that $\pi \circ s = \mathrm{id}_X$ is called a (global) **section** (if $s : U \subseteq X \to E$ we call it a local section). We denote the space of global sections by $\Gamma(E)$ and by $\Gamma(U)$ the space of local sections over an open subset $U \subseteq X$.

def:vb-homomorphismOfVectorBundles

Definition 2.1.12 (Homomorphism of Vector Bundles)**.** A **homomorphism between two vector bundles** $\pi : E \to X$ and $\tilde{\pi} : F \to X$ over the same base space is a continuous function $\varphi : E \to F$ such that:

1. $$\begin{tikzcd} E \arrow[r, "\varphi"] \arrow[d, "\pi"] & F \arrow[dl, "\tilde{\pi}"] \\ X \end{tikzcd}$$ commutes.
2. For all $x \in X$ the function $\varphi_x : E_x \to F_x$ is linear.

def:vb-pullbackOfVerctorBundles

Definition 2.1.13 (Pullback of Vector Bundles)**.** Let $\pi : E \to Y$ be a vector bundle, X be a topological space and $f : X \to Y$ be continuous. Then we define the **pullback** $f^*\pi : f^*(E) \to X$ as follows: The total space $f^*(E)$ is defined as the subspace of $(x, e) \in X \times E$ such that $f(x) = \pi(e)$, and the map $f^*\pi$ is the projection to the second factor restricted to said subspace.

Example 2.1.14. Let $\pi : E \to X$ be a vector bundle. For any $Y \subseteq X$ we have the vector bundle
$$\pi|_Y : \pi^{-1}(Y) \to Y\,,$$
which is defined as the pullback along the inclusion of Y into X.

rem:vb-restrictionOfBundles

Remark 2.1.15. We will often omit the notation of the projection and denote a vector bundle by its total space. The restriction $\pi|_Y : \pi^{-1}(Y) \to Y$ is often denoted by $E|_Y$. The restriction may equivalently be defined by restricting the domain and codomain of the projection.

rem:vg-pullbackInducesBundleMap

Remark 2.1.16. Notice that this construction of the pullback $f^*\pi : f^*(E) \to X$ is the simplest space to make the square commute:

$$\begin{tikzcd} f^*(E) \arrow[r, "\pi_2"] \arrow[d, "f^*\pi"] & E \arrow[d, "\pi"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

The notion of simplicity is captured by the universal property, that for any other vector bundle $\tilde{\pi} : \tilde{E} \to X$ and function $\tilde{f} : \tilde{E} \to E$ there exists a unique $h : \tilde{E} \to f^*(E)$ (which is given by $h(y) := (\tilde{\pi}(y), \tilde{f}(y))$ such that the following diagram commutes:

$$\begin{tikzcd}
\tilde{E} \arrow[dr, "h"] \arrow[drr, bend left, "\tilde{f}"] \arrow[ddr, bend right, "\tilde{\pi}"'] & & \\
& f^*(E) \arrow[r, "\pi_2"] \arrow[d, "f^*\pi"] & E \arrow[d, "\pi"] \\
& X \arrow[r, "f"] & Y
\end{tikzcd}$$

Remark 2.1.17. Let $\pi : E \to X$ be a vector bundle. Then the space of sections $\Gamma(E)$ is a (in general, infinite-dimensional) $\mathbb{R}$-vector space. Furthermore, and more relevant to the purposes of this thesis, we view $\Gamma(E)$ (and $\Gamma(U)$ for open subsets) as a $C(X)$ (or $C(U)$) module. The latter denotes the ring of continuous functions from E (or U) to $\mathbb{R}$.

Example 2.1.18. For any smooth manifold M of dimension m , we can equip the union

$$TM := \bigsqcup_{p \in M} T_pM$$

with a topology, such that $\pi : TM \to M$ defines a vector bundle which we call **tangent bundle**. For this we first define the trivialisations and then equip TM with a topology to make them homeomorphisms: Let $\varphi : U \to V$ be a chart on M. If we denote the i-th coordinate function of φ by φ_i we can define the bijection

$$\tilde{\varphi} : \pi^{-1}(U) \to V \times \mathbb{R}^m, \quad \xi_p \mapsto \left(\varphi(p), \begin{pmatrix} \xi_p(\varphi_1) \\ \vdots \\ \xi_p(\varphi_m) \end{pmatrix} \right).$$

Now we declare a subset $W \subseteq TM$ to be open, if and only if for all charts φ the set $\tilde{\varphi}(W \cap \pi^{-1}(U)) \subset V \times \mathbb{R}^m$ is open. We can now extend and give meaning to the notation of the local basis vectors $\partial_i(p) \in T_pM$: For any chart $\varphi : U \to V$ we have the local sections

$$\partial_i : U \to TU, \quad p \mapsto \partial_i(p).$$

Furthermore, TM is a $2m$-dimensional manifold. Using a smooth atlas on M yields a smooth atlas on TM making it a smooth manifold. Notice that the family $(\partial_i)_i$ is a basis of the $C(U)$-module $\Gamma(U)$. This follows by expressing any section in local coordinates via $\tilde{\varphi}$.

Notice that TM as a bundle satisfies more then just the bundle definition: Taking two different charts induces two different local trivialisations. Transitioning between them is not just fibrewise linear, but also smooth. This is captured in the definition:

Henceforth we suppress the differentiable structure from the notation.

Definition 2.1.19 (The Derivative)**.** Let M and M' be smooth manifolds. We call a continuous function $f : M \to M'$ **smooth**, if for all $\varphi \in \mathfrak{A} \in c$ and $\varphi' \in \mathfrak{A}' \in c'$ the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \to V'$$

are smooth. A given smooth function induces a smooth map between the tangent bundles as manifolds as follows:

$$\mathrm{d}f : TM \to TM' \ , \mathrm{d}f_p(\xi_p)(g_p) = \xi((g \circ f)_p)$$

Here, $\xi_p \in T_pM$, $g_p \in \mathcal{E}_p(M')$. The function $\mathrm{d}f$ is called the **derivative** or **pushforward** of f.

Definition 2.1.20 (Smooth Vector Bundles)**.** A **smooth vector bundle of rank** r is a surjection $\pi : E \to M$ where the codomain (or base space) is a smooth manifold together with a family of continuous functions (or local trivialisations)

$$\Phi_i : \pi{-}1(U_i) \to U_i \times \mathbb{R}^r$$

such that $\bigcup_i U_i = M$ and for all overlapping $U_i \cap U_j \neq \varnothing$, the transition functions

$$\Phi_j \circ \Phi_i^{-1} : \pi^{-1}(U_i \cap U_j) \to (U_i \cap U_j) \times \mathbb{R}^r$$

are of the form

$$\Phi_j \circ \Phi_i^{-1}(p, v) = (p, g_{i,j}(v)), \quad \text{where} \quad g_{i,j} : (U_i \cap U_j) \to \mathrm{GL}(r, \mathbb{R})$$

is smooth.

Corollary 2.1.21. A smooth vector bundle is a smooth manifold and also a vector bundle. The latter is ensured by the codomain of $g_{i,j}$ which is also called the **structure group**. We can enforce structure on the bundle by reducing the structure group. As a preview: reducing it to $\mathrm{GL}^+(k, \mathbb{R})$ the linear maps with positive determinant means that we can induce the concept of orientations. Reducing it to $\mathrm{O}(k)$ lets us equip it with a Riemannian metric.

def:vb-framesFromTrivialisations

Definition 2.1.22. Let $\pi : E \to M$ be a smooth vector bundle of rank r and let

$$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^r$$

be a local trivialisation. Then a section is called smooth if it is a smooth map of manifolds. A family of smooth local sections $(s_1, \ldots, s_r)$ with $s_i \in \Gamma(U)$ is called a **local frame** if they form a basis of the $C(U)$-module $\Gamma(U)$. Such a family is obtained from Φ by defining $s_i(p) := \Phi^{-1}(p, e_i)$. Similar, from a local frame we can obtain a local trivialisations by defining:

$$\Phi : \pi^{-1}(U) \to U \times \mathbb{R}^r \, , \quad v = \sum_{i=1}^{r} \lambda_i s_i(\pi(v)) \mapsto (\pi(v), \lambda_1, \ldots, \lambda_r) \, .$$

Example 2.1.23. The local trivialisations ∂_i of the tangent bundle of a smooth manifold M form a local frame and make the tangent bundle into a smooth vector bundle.

Definition 2.1.24 (Compact-open topology)**.** Let X, Y be topological spaces. Then the set of continuous maps $C(X,Y)$ can be equipped with the topology that has

$$\{V(K,U) | K \subseteq X \text{ compact }, U \subseteq Y \text{ open}\} \text{ where } V(K,U) \coloneqq \{f \in C(X,Y) |\ f(K) \subseteq U\}$$

as a subbasis. This coincides with the topology of $\mathrm{Hom}(V,W)$ as a subset of $\mathbb{C}^n$ after identification with the corresponding matrix.

def:vb-homomorphismBundle

Definition 2.1.25 (Homomorphism Bundle)**.** Let W, V be vector spaces and $E = V \times X$, $F = W \times X$ the product bundles over X. Now any bundle homomorphism $\varphi : E \to F$ determines a map $\Phi : X \to \hom(V,W)$ by sending $x \mapsto \varphi_x$. With respect to the compact-open topology, this Φ is continuous. Furthermore, a continuous map $\Phi : X \to \hom(V,W)$ determines a bundle homomorphism $\varphi : E \to F$.

Proof. Let $\{e_i\}$ and $\{f_i\}$ be bases of V and W. Then each $\Phi(x)$ is a matrix $\Phi_{i,j}(x)$, such that:

$$\Phi(x)e_i = \sum_j \Phi(x)_{i,j} f_j \,.$$

Now Φ is continuous if and only if $\Phi_{i,j}$ is continuous. Since $\varphi(x,v) = (x, \Phi(x)v)$ this is equivalent to φ being continuous. □

thm:vb-isomorphicVectorBundles

Lemma 2.1.26 (Isomorphic Vector Bundles)**.** *Let E and F be vector bundles and $\varphi : E \to F$ be a homomorphism. Then φ is an isomorphism if and only if all φ_x are isomorphisms.*

Proof. To show that it is an isomorphism we need an continuous inverse, equivalently, we need to show that φ is a homeomorphism. First of all we can deduce, that φ is bijective: Assume it wasn't injective, then there would be a fibre in which φ_x is not injective. Assume that it wasn't surjective. Then there would be a $w \in F$ that has no preimage which means that in the fiber containing w there is no isomorphism. Thus, there is a unique inverse and it remains to verify that it is continuous, which is a local condition. Therefore we can assume that E and F are product bundles. Then we have a continuous map $\Phi : X \to \mathrm{Hom}(V,W)$. Furthermore, since all φ_x are isomorphisms, we can restrict the image to $\mathrm{Iso}(V,W)$ which is an open subset of $\mathrm{Hom}(V,W)$. Now we can define the map $x \mapsto \Phi(x)^{-1}$ which is continuous since taking inverses is continuous. By the corollary above, we obtain a continuous map $\psi : F \to E$ by sending $(x,w) \mapsto (x, \Phi(x)^{-1}(w))$, which is the inverse. □

Example 2.1.27. Let M be a smooth manifold of dimension m, $\varphi : U \to V$ a chart and $\tilde{\varphi} : TU \to V \times \mathbb{R}^m$ be the corresponding bundle chart. Then it follows from the lemma, that TU is isomorphic to $V \times \mathbb{R}^m$. This is verified by observing that $\partial_i(p)$ is a basis of T_pU for all $p \in U$. In this case $\tilde{\varphi}\,(\partial_i) = e_i$ giving us isomorphisms in each fiber.

Now we want to construct new bundles from old ones. There are many operations to construct new vector spaces such as the tensor product, direct sum, Hom-Spaces and many more. Instead of individually proving that those operations applied fiberwise yield new bundles, we subsume these operations within the following definition:

Definition 2.1.28 (Continuous Functor)**.** Let T be a covariant endofunctor in the category of finite-dimensional vector spaces. We call T a **continuous functor**, if for all V, W the map $T : \hom(V,W) \to \hom(T(V),T(W))$ is continuous. All vector spaces are given the topology induced by their isomorphism to real Euclidean space. If E is a vector bundle we define the set:

$$T(E) \coloneqq \bigcup_{x\in X} T(E_x)\,.$$

Now we want to equip this with a topology such that $T(E)$ is a bundle over the same base space as E.

def:vb-constructingNewBundles

Definition 2.1.29 (Constructing new bundles)**.** We start by assuming that $E = X \times V$. In that case we define the topology on $T(E)$ as the product topology $X \times T(V)$. Now suppose that $F = X \times W$ is another product bundle and $f : E \to F$ is a homomorphism. Let $\Phi : X \to \hom(V,W)$ be the corresponding map. Then $T\Phi : X \to \hom(T(V),T(W))$ is continuous by assumption. Thereby, $T(\varphi) : X \times T(V) \to X \times T(W)$ is continuous. If φ is an isomorphism, then $T(\varphi)$ is also an isomorphism. That follows because it is continuous and an isomorphism fibrewise.

Now assume that E is trivial. Then we can choose an isomorphism $\alpha : E \to X \times V$ and induce a topology on $T(E)$ by pulling it back via $T(\alpha) : T(E) \to X \times T(V)$. This topology does not depend on α by the above. For general vector bundles we do this construction locally.

If $f : X \to Y$ is continuous, there is a natural isomorphism

$$f^*T(E) \cong Tf^*(E)$$

This construction is carried out in detail by Atiyah in [atiyah1989k Ati89, Chapter 1.2] and can be generalised for continuous functors with several variables both co- and contravariant. Hence this procedure gives us for vector bundles E and F over the same base space the new bundles:

(i) $E \oplus F$, their direct sum

(ii) $E \otimes F$, their tensor product

(iii) $\mathrm{Hom}(E,F)$

(iv) E^*, the dual bundle of E

(v) $\Lambda^i(E)$ the i-th exterior power

and by the natural isomorphism we have natural isomorphisms:

(i) $E \oplus F \cong F \oplus E$

(ii) $E \otimes F \cong F \otimes E$

(iii) $E \otimes (F \oplus F') \cong E \otimes F \oplus E \otimes F'$

(iv) $\mathrm{Hom}(E, F) \cong E^* \otimes F$

cor:vb-sectionsIntoHom

Corollary 2.1.30. Let $\pi_E : E \to X$ and $\pi_F : F \to X$ be two vector bundles over X. Then, the space $\Gamma(\mathrm{Hom}(E, F))$ can be identified with the space of bundle morphisms $\mathrm{hom}(E, F)$ since for any section $s : X \to \mathrm{Hom}(E, F)$ we can define the bundle homomorphism $\tilde{s} : E \to F$ by sending $v \mapsto s_{\pi_E(v)}(v)$. This is a homomorphism since $\pi_F(\tilde{s}(v)) = \pi_E(v)$.

Example 2.1.31 (Cotangent bundle)**.** Let M be a smooth manifold of dimension m. Then we can define the **cotangent bundle** T^*M to be the dual of the tangent bundle. A local frame of said bundle is given by the fiberwise dual of the frame (∂_i). Sections into the tangent bundle are called **vector fields** and sections into the cotangent bundle are called **one-forms**.

def:vb-linearIdentification

Definition 2.1.32. Let V be a linear manifold (i.e. a finite-dimensional vector space). Then for any $v \in V$ we have the canonical maps

$$\begin{aligned} R_v : V &\to T_v V \\ x &\mapsto \left(f_v \mapsto \frac{\partial}{\partial t}\Big|_{t=0} f(v + tx) \right) \end{aligned}$$

Let $L : V \to V$ be a linear map viewed as a smooth map between manifolds. Then we have the identity:

$$\mathrm{d}L_v \circ R_v = R_{L(v)} \circ L\,.$$

This is proved by a short calculation:

$$\Big(\mathrm{d}L_v\big(R_v(x)\big)\Big)(f_{L(v)}) = R_v(x)\,(f \circ L) = \frac{\partial}{\partial t}\Big|_{t=0} f \circ L(v + tx) = \underbrace{\frac{\partial}{\partial t}\Big|_{t=0} f\big(L(v) + tL(x)\big)}_{=R_{L(v)} \circ L(x)}$$

Now let $v, v' \in V$. Then we have the translation

$$tr_{v,v'} : V \to V; \quad x \mapsto (x + (v' - v))$$

This allows us to relate R_v and $R_{v'}$ as follows:

$$\left(\mathrm{d}tr_{v,v'}\big|_v\,(R_v(x))\right)(f_{v'}) = \frac{\partial}{\partial t}\Big|_{t=0} (f \circ tr_{v,v'}(v + tx)) = \frac{\partial}{\partial t}\Big|_{t=0} \big(f(v' + tx)\big) = R_{v'}(x)(f_{v'})\,.$$

Hence:

$$R_{v'} = \mathrm{d}tr_{v,v'} \circ R_v\,.$$

The **linear identification** is the map Q given on the whole tangent bundle:

$$Q : TV \to V; \quad \xi_p \mapsto R_p^{-1}(\xi_p)\,,$$

and satisfies for any linear $L : v \to V$

$$Q \circ \mathrm{d}L = L \circ Q\,.$$

cor:vb-vectorFieldsTakeFunctions

Corollary 2.1.33. Let $f : M \to \mathbb{R}$ be smooth. Then we can interpret $\mathrm{d}f$ as a one-form. To be precise, let $Q : T\mathbb{R} \to \mathbb{R}$ denote the linear identification with respect to the identity on $\mathbb{R}$. For $v \in \Gamma(TM)$ we have

$$(Q \circ \mathrm{d}f)(v) = v(f)$$

Here on the right side, f denotes a map $p \mapsto f_p$ such that $v(f)(p) := v_p(f_p)$ is well defined. We retain this notation so that vector fields can take functions as an input.

Proof. Let $\varphi : U \to V$ be a chart of M. We prove this by showing it for the local sections ∂_i and the general case follows by $C(U)$-linearity, since those sections form a local frame. Now let $g_p \in \mathcal{E}_p(\mathbb{R})$ and $\varphi(p) = x_0$. Then

$$\mathrm{d}f_p(\partial_i(p))(g_p) = \partial_i((g \circ f)_p) == \frac{\partial}{\partial x_i}\Big|_{x_0} (g \circ f \circ \varphi^{-1}) = \frac{\partial}{\partial x}_{f_p(p)} (g_p) \cdot \partial_i(p)(f_p)$$

Hence, applying Q to the calculation yields $(Q \circ \mathrm{d}f)(v) = v(f)\,.$ $\square$

Definition 2.1.34 (A Metric)**.** Let M be a smooth manifold. A section $g \in \Gamma(T^*M \otimes T^*M)$ into the tensor product of the dual tangent bundle with itself is a **Riemannian metric** if the following are satisfied for all $p \in M$:

1. g is **non-degenerate**, meaning if for a $v_p \in T_pM$ we have $g_p(v_p, w_p) = 0$ for all $w_p \in T_pM$ then $v_p = 0$.

2. g is **symmetric**, meaning $g_p(v_p, w_p) = g_p(w_p, v_p)$ for all $v_p, w_p \in T_pM$.

3. g is **positive definite**, meaning that $g_p(v_p, v_p) \geq 0$ for all v_p with equality only for $v_p = 0$.

We call a tuple (M, g) a **Riemannian manifold**.

Remark 2.1.35. Equipping a smooth manifold M of dimension m with a Riemannian metric is equivalent to reducing the structure group of the tangent bundle from $\mathrm{GL}(m)$ to $\mathrm{O}(m)$. To be precise: A Riemannian metric g lets us canonically choose an atlas with structure group $\mathrm{O}(m)$ and an atlas with such a structure group lets us canonically choose a Riemannian metric.

Proof. We start by assuming that we have an atlas with structure group $\mathrm{O}(m)$. In Definition 2.1.22 we saw that a covering of local trivializations is equivalent to a covering of smooth frames. Now in a single trivialization

$$\Phi : \pi^{-1}(U) =: TU \to U \times \mathbb{R}^m$$

we call the projection to the $\mathbb{R}^m$-part (the second factor) π_2 and define

$$\begin{aligned} g : U &\to T^*U \otimes T^*U \\ p &\mapsto \begin{array}{ll} g_p^{\Phi} : T_pU \times T_pU & \to \mathbb{R} \\ \qquad (\nu_p, \xi_p) & \mapsto g_p^{\Phi}(\nu_p, \xi_p) := \pi_2\left(\Phi(\nu_p)\right) \cdot \pi_2\left(\Phi(\xi_p)\right) , \end{array} \end{aligned}$$

where $\cdot$ denotes the standard scalar product. This gives us locally a Riemannian metric. If we do this for all trivializations we have to make sure, that the result is well defined. So assume that two trivializations Φ and Ψ overlap in V. The structure group is $\mathrm{O}(m)$, $\Phi = R\Psi$ for some function

$$R : V \to \mathrm{O}(m) .$$

Let p be in said overlapping and (ν_p, ξ_p) be from $T_pM \times T_pM$. Then since R is orthogonal we can calculate the independence from the trivialization:

$$g_p^{\Phi}(\nu_p, \xi_p) = \pi_2\left(R\Psi(\nu_p)\right) \cdot \pi_2\left(R\Psi(\xi_p)\right) = \pi_2\left(\Psi(\nu_p)\right) \cdot \pi_2\left(\Psi(\xi_p)\right) = g_p^{\Psi}(\nu_p, \xi_p) .$$

To get an orthonormal atlas from a smooth one given a Riemannian metric, we start by building a covering of local frames from the trivialisations. We then apply Gram-Schmidt to the local frames to get orthonormal frames. Now a covering of orthonormal frames gives a covering of trivializations, where the structure group consists of basis changes between orthonormal bases and hence is $\mathrm{O}(m)$. □

Definition 2.1.36 (The Gradient)**.** Assume that (M, g) is a smooth manifold together with a Riemannian metric. Let $f : M \to \mathbb{R}$ be a smooth function and $Q : T\mathbb{R} \to \mathbb{R}$ the linear identification. We define its **gradient** to be a vector field $\mathrm{grad}(f) \in \Gamma(TM)$ such that for any vector field V:

$$Q \circ \mathrm{d}f(V) = g(\mathrm{grad}(f), V) .$$

Notice how the non-degeneracy of g ensures existence and uniqueness.

def:vb-localDescriptionOfGradientGij

Definition 2.1.37. Let (M, g) be a Riemannian manifold and $\varphi : U \to V$ be a chart. We define the smooth functions $g_{ij} : U \to \mathbb{R}$ to be:

$$g_{ij}(p) = g_p(\partial_i(p), \partial_j(p)) .$$

Then if two vector fields v, w over U are given with $v = \sum v_i \partial_i$ and $w = \sum_i w_i \partial_i$ we can calculate the metric locally:

$$g(v, w) = \sum_{i,j} g_{ij} v_i w_j \ : U \to \mathbb{R}$$

Furthermore, we can invert the matrices $(g_{ij}(p))_{ij}$ for all p (which is a smooth procedure as can be seen by the construction via Cramer's rule) to get smooth functions

$$\begin{aligned} g^{ij} : U &\to \mathbb{R} \\ p &\mapsto g^{ij}(p) = \left((g_{kl}(p))^{-1}_{kl} \right)_{ij} \end{aligned}$$

Remark 2.1.38. Due to the rare appearance of big summations with vectors and co-vectors in this work, we omit the Einstein summation convention and assign all coordinate entries lower indices.

lem:vb-localDescriptionOfGradient

Lemma 2.1.39 (Local Description of the Gradient)**.** *Let (M, g) be a smooth Riemannian manifold of dimension m and $\varphi : U \to V$ be a chart. Then the gradient has the local form:*

$$\mathrm{grad}(f) = \sum_{i,j} g^{ij} \left(\partial_i(f)\right) \cdot \partial_j \,.$$

Proof. Assume that $v, w \in \Gamma(TU)$ such that $v = \sum_i v_i \partial_i$ and $w = \sum_i w_i \partial_i$ (Here v_i and w_i are smooth functions $U \to \mathbb{R}$). Then $g(v, w) = \sum_{i,j} g_{ij} v_i w_j$. Suppose that $\mathrm{grad}(f) = \sum_j G^j \partial_j$ and Q is the linear identification $T\mathbb{R} \to \mathbb{R}$ in each tangent space. Then by definition of the gradient and corollary 2.1.33 we have:

$$\partial_j(f) = Q \circ \mathrm{d}f(\partial_j) = g(\mathrm{grad}(f), \partial_j) = g(\sum_i G^i \partial_i, \partial_j) = \sum_i g_{ij} G^i$$

Reading this as a pointwise matrix product:

$$\begin{aligned} (G^1, \cdots G^m)(g_{ij}) &= (\partial_1(f), \cdots, \partial_m(f)) \\ \text{which gives } (G^1, \cdots G^m) &= (\partial_1(f), \cdots, \partial_m(f))\,(g^{ij})\,. \end{aligned}$$

But then we can conclude that $G^j = \sum_i g^{ij} \partial_i(f)$.

□

Definition 2.1.40 (A Dynamical System)**.** Let M be a smooth manifold. A **dynamical system on** M (sometimes also called a **flow on** M) is given by a smooth map $\varphi : \Omega \to M$ such that:

1. $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$ open, and for any $p \in M$ the set $I(p) \coloneqq \pi_1(\Omega \cap \mathbb{R} \times \{p\})$ is connected. (π_1 is the projection onto the first factor)
2. For all $p \in M$ and $t \in I(p)$ we have that $s \in I(\varphi(t, p))$ if and only if $s + t \in I(p)$. Then we have:

$$\varphi(s, (\varphi(t, p))) = \varphi(s + t, p)$$

3. For all p we require $\varphi(0, p) = p$.

We will write $\varphi_t(p) := \varphi(t, p)$ to keep the notation manageable. If for all p $I(p) = \mathbb{R}$ we call the system **global**.

Corollary 2.1.41. For a global dynamical system φ on a smooth manifold M the map

$$\rho : \mathbb{R} \to \mathrm{Diff}(M), \quad t \mapsto \varphi_t$$

is a homomorphism. To verify well-definedness notice how for all $t \in \mathbb{R}$, φ_{-t} is an inverse of φ_t. The map ρ is called a **one-parameter group of diffeomorphisms**.

Definition 2.1.42 (Vector Field of a Dynamical System)**.** Let $\varphi : \Omega \to M$ be a dynamical system on a smooth manifold M. For a chart $\psi : U \to V$ of M we have the chart $\mathrm{id} \times \psi$ of Ω. For this chart we denote the associated local frame by $(\partial_t, \partial_1, \ldots, \partial_m)$. Using this we define the mapping

$$\begin{aligned} M &\to TM \\ p &\mapsto X(p) := \mathrm{d}\varphi(\partial_t(0, p)) \end{aligned}$$

and call it the **associated vector field**. This is in fact a smooth vector field.

Definition 2.1.43 (Dynamical System of a Vector Field)**.** Let $X \in \Gamma(TM)$ be a global vector field on the smooth manifold M. Then we call a dynamical system φ on M the **associated dynamical system** to X, if for all $t \in \mathbb{R}$ and $x \in M$

$$\mathrm{d}\varphi\left(\partial_t(t, x)\right) = X(\varphi_t(x)) .$$

Remark 2.1.44. Regarding the left-hand side of the definition $\mathrm{d}\varphi\left(\partial_t(t, x)\right)$ we could also differentiate the map

$$t \mapsto \varphi_t(x) ,$$

for fixed x at t and evaluate this at the constant vector field $t \mapsto R_t(1) \in T_t\mathbb{R}$ from definition 2.1.32.

These two constructions are mutually inverse, giving the following theorem:

Theorem 2.1.45. *Let M be a smooth manifold. Then there is a one-to-one correspondence between global dynamical systems and vector fields given by the constructions above.*

3 Complex K Theory

3.1 Mathematical Goals

Theorem 3.1.1 (The Theorems and Mathematical Structures). • *Definition of The K- Functor*

- *Cohomology structures*
  - *Additivity (pointed sum)*
  - *Long exact sequences*
  - *Excision*

3.2 Classification of Vector Bundles

The first goal is to classify all vector bundles over sufficiently nice spaces. The latter refers to spaces admitting partitions of unity for which Tietze's extension theorem holds. For simplicity we just consider compact Hausdorff spaces. Along the way we will discover that the set of isomorphism classes of vector bundles is a homotopy invariance.

Before we go back to the analytical reals, we will further develop the theory of vector bundles:

Definition 3.2.1. For a topological space X we define Vect_n to be the set of isomorphism classes of complex vector bundles of rank n and $\mathrm{Vect}(X)$ to be the set of isomorphism classes of complex vector bundles of any rank. By declaring $\mathrm{Vect}(f) = f^*$ to be the pullback we get a contravariant functor Vect from **Top** to **Set**. We can equip the set $\mathrm{Vect}(X)$ with the direct sum and tensor product operation. It is then easy to see that these operations descend to isomorphism classes and hence we get an abelian semi ring $(\mathrm{Vect}(X), \oplus, \otimes)$ (abelian semigroup with associative and distributive multiplication). In fact we even get a contravariant functor from **Top** to the category of abelian semirings.

lem:kTheo-sectionExtension

Lemma 3.2.2. *Let X be a compact Hausdorff space, $Y \subseteq X$ a closed subspace and E a vector bundle over X. Then any section $s : Y \to E|_Y$ can be extended to a global section of X.*

Proof. For a detailed proof see Chapter 1.4 of [Ati89] (atiyah1989k). The general argument however is easy to follow: By Tietze's extension theorem we can extend the section to around every $x \in X$. By compactness we can pick finitely many of those extensions. Now using a partition of unity with respect to their domains we can build the sum of all the extensions to get a well defined global one. □

cor:kTheo-isomorphismExtension

Corollary 3.2.3. Let Y be a closed subspace of a compact Hausdorff space X, and let E, F be bundles over X. If $f : E|_Y \to F|_Y$ is an isomorphism, then there is an open neighbourhood $U \supset Y$ and an extension $f : E|_U \to F|_U$ which is an isomorphism.

Proof. By corollary 2.1.30 we can view f as a section into the bundle $\mathrm{Hom}(E|_Y, F|_Y)$, which can be extended to a global one in $\mathrm{Hom}(E, F)$. Let U be the open set for which the section maps to isomorphisms. Since U contains Y this yields the require extension. □

thm:kTheo-homotopyInvariance

Theorem 3.2.4 (Homotopy Invariance of the Pullback)**.** *Let X, Y be compact Hausdorff spaces, $I = [0, 1]$, and*

$$f : Y \times I \to X$$

be a homotopy. For a fixed $t \in I$, we define the function $f_t : Y \to X$, $f_t(y) = f(t, y)$. Let $\pi : E \to X$ be a vector bundle. Then

$$f_0^*(E) \cong f_1^*(E)\,.$$

Proof. Consider the following two vector bundles over $Y \times I$:

$$f^*E \quad \text{and} \quad \pi_1^* f_t^* E\,,$$

where $\pi_1 : Y \times I \to Y$ denotes the projection and $t \in I$. Since the restriction is by definition the pullback along the inclusion $i : Y \times \{t\} \to Y \times I$. The equality of the two maps:

$$f \circ i = f_t \circ \pi_1 \circ i : Y \times \{t\} \to X$$

together with the functoriality gives us an isomorphism

$$\Phi : f^*E|_{Y\times\{t\}} \to \pi_1^* f_t^* E|_{Y\times\{t\}}\,.$$

By Corollary 3.2.3 we can extend the isomorphism for some open neighbourhood of $Y \times \{t\}$ and by the product topology we can deduce that there exists a δ_t such that U contains the thickened line $Y \times (t - \delta_t, t + \delta_t)$. The isomorphism class of $f^*E|_{Y\times\{t\}}$ as a bundle over Y is locally constant in t: Let $0 < \varepsilon < \delta$ and $r \in I$. Then define the inclusions

$$\begin{aligned} i_t : Y &\to Y \times \{t\} \subseteq Y \times I \\ y &\mapsto (y, t) \end{aligned} \qquad \text{and} \qquad \begin{aligned} i_{t+\varepsilon} : Y &\to Y \times (t - \delta_t, t + \delta_t) \\ y &\mapsto (y, t + \varepsilon) \end{aligned}$$

Then we have the identities $f_t = f \circ i_t$ and $\pi_1 \circ i_{t+\varepsilon} = \mathrm{id}_Y$ and we can calculate:

$$f_{t+\varepsilon}^* E = i_{t+\varepsilon}^* f^* E = i_{t+\varepsilon}^* \pi_1^* f_t^* E = (f_t \circ \pi_1 \circ i_{t+\varepsilon})^* E = f_t^* E\,.$$

Finally, since I is connected, any locally constant map from I is globally constant and therefore $f_0^* E \cong f_1^* E$. □

lem:kTheo-contractibleBundlesAreTivial

Corollary 3.2.5.
1. If $f : X \to Y$ is a homotopy equivalence, $f^* : (Y) \to (X)$ is a semiring isomorphism.
2. If X is contractible, every bundle over X is trivial and $(X) \cong \mathbb{N}_0$

def:kTheo-trivialization

Definition 3.2.6 (Trivialization of a Vector Bundle)**.** Let Y be a closed subspace of X, E a vector bundle over X and $\alpha : E|_Y \to Y \times V$ be an isomorphism. Then we call α **trivialisation of E over Y**.

lem:kTheo-trivialisationDefinesABundle

Lemma 3.2.7. *A trivialization α of a bundle E over $Y \subseteq X$ defines a bundle E/α over X/Y. The isomorphism class of E/α depends only on the homotopy class of α.*

Proof. We define the bundle E/α as follows: Let $\pi_2 : Y \times V \to V$ denote the projection and define an equivalence relation on $E|_Y$ by

$$e \sim e' \;\Leftrightarrow \pi_2\alpha(e) = \pi_2\alpha(e') \,.$$

We extend this relation by the identity on $E|_{X \setminus Y}$ and we define E/α to denote the quotient space of E defined by that relation. To see why this is a bundle we need to check the local triviality only around the base point Y/Y of X/Y. This follows from the fact that by Corollary 3.2.3 we can extend α to an isomorphism $\tilde{\alpha} : E|_U \to U \times V$. Then $\tilde{\alpha}$ induces an isomorphism

$$E/\alpha\big|_{U/Y} \cong (U/Y) \times V$$

proving the local triviality. Now for the homotopy invariance of said bundle suppose that α_0 and α_1 are homotopic trivializations of E over Y. This means that we have a trivialization β of $E \times I$ over $Y \times I \subset X \times I$ that induces the α_i at the endpoints. Let $f : (X/Y) \times I \to (X \times I)/(Y \times I)$ be the natural map. Then $f^*((E \times I)/\beta)$ is a bundle on $(X/Y) \times I$ whose restriction to $(X/Y) \times \{i\}$ is E/α_i $(i = 0, 1)$. Hence by the homotopy invariance of the pullback (Theorem 3.2.4) we have

$$E/\alpha_0 \cong E/\alpha_1 \,.$$

□

lem:kTheo-collapsingOfContractibleSubspace

Lemma 3.2.8. *Let $Y \subseteq X$ be a closed contractible subspace of a compact Hausdorff space X. Then*

$$f : X \to X/Y \quad \text{induces a semiring isomorphism} \quad f^* : \mathrm{Vect}(X/Y) \to \mathrm{Vect}(X) \,.$$

Proof. Since Y is contractible, $E|_Y$ is trivial and thereby a trivialization $\alpha : E|_Y \to Y \times V$ exists. Two such trivializations only differ by the automorphism of $Y \times V$ which in turn corresponds to a map $Y \to \mathrm{GL}(V) \cong \mathrm{GL}_{\mathrm{rank}(E)}(\mathbb{C})$. Now since Y is contractible those two maps factor (up to homotopy) over any one point $y \in Y$ and are thereby homotopically equivalent to the constant map. Finally since $\mathrm{GL}_{\mathrm{rank}(E)}(\mathbb{C})$ is connected all constant maps are homotopically equivalent. Thereby α is unique up to homotopy. By Lemma 3.2.7 the isomorphism class of E/α only depends on the homotopy class of α we have a well defined map

$$\mathrm{Vect}(X) \to \mathrm{Vect}(X/Y) \,,$$

which in turn is an inverse of f^* concluding the proof. □

def:kTheo-gluingAndClutching

Definition 3.2.9 (Gluing and Clutching of Vector Bundles)**.** Let

$$X = X_1 \cup X_2\,, \quad A = X_1 \cap X_2\,,$$

and all be compact Hausdorff spaces. Assume that E_i is a vector bundle over X_i and $\varphi : E_1|_A \to E_2|_A$ is an isomorphism. Then we define a vector bundle $E_1 \cap_\varphi E_2$ on X as follows: We topologize the space $E_1 \cup_\varphi E_2$ as the quotient of $E_1 + E_2/\sim$, where $\sim$ is the identification of points with their image under φ. Identifying X with $X_1 + X_2$ in the quotient space we get the natural projection $p : E_1 \cap_\varphi E_2 \to X$ and the fibers have a natural vector space structure. It remains to show that this is locally trivial. This is obvious outside of A so let $a \in A$ and V_1 be a closed neighbourhood of a in X_1 such that $E_1|_{V_1}$ is trivial giving us the isomorphisms

$$\theta_1 : E_1|_{V_1} \to V_1 \times \mathbb{C}^n$$
$$\theta_1^A : E_1|_{V_1 \cap A} \to (V_1 \cap A) \times \mathbb{C}^n$$

Finally, define

$$\theta_2^A : E_2|_{V_1 \cap A} \to (V_1 \cap A) \times \mathbb{C}^n$$

by composition $\theta_1^A \circ \varphi^{-1}$. Let V_2 be a neighborhood of a in X_2 such that

$$\theta_2 : E_2|_{V_2} \to V_2 \times \mathbb{C}^n$$

is an extension of θ_2^A. This extension exists again by the extending the section into the homomorphism bundle corresponding to the isomorphism θ_2^A. Finally θ_1 and θ_2 defines a well defined isomorphism

$$\theta_1 \cap_\varphi \theta_2 : E_1 \cap_\varphi E_2 \to (V_1 \cap V_2) \times \mathbb{C}^n\,.$$

This proves the local triviality.

We call such a triple (E_1, E_2, φ) a **clutching datum** and φ the **transition function**.

cor:kTheo-elementaryPropertiesGluingAndClutching

Corollary 3.2.10 (Elementary Properties of the Gluing and Clutching Construction)**.** Elementary properties of the gluing and clutching construction are:

1. If E is a bundle over X, then the identity defines ans isomorphism and
$$E_1 \cup_{\mathrm{id}_A} E_2 \cong E.$$
2. If $\beta_i : E_i \to E_i'$ are isomorphisms on X_i and $\varphi'\beta_1 = \beta_2\varphi$, then
$$E_1 \cup_\varphi E_2 \cong E_1' \cup_{\varphi'} E_2'\,.$$
3. If (E_i, φ) and (E_i', φ') are two "clutching data" on X_i, then
$$(E_1 \cup_\varphi E_2) \oplus (E_1' \cup_{\varphi'} E_2') \cong E_1 \oplus E_1' \bigcup_{\varphi \oplus \varphi'} E_2 \oplus E_2'\,,$$
$$(E_1 \cup_\varphi E_2) \otimes (E_1' \cup_{\varphi'} E_2') \cong E_1 \otimes E_1' \bigcup_{\varphi \otimes \varphi'} E_2 \otimes E_2'\,,$$
$$(E_1 \cup_\varphi E_2)^* \cong E_1^* \bigcup_{(\varphi^*)^{-1}} E_2^*\,.$$

$(-)^*$ denotes the dual here.

lem:kTheo-homotopyInvarianceClutchingFunction

Lemma 3.2.11. *The isomorphism class of $E_1 \cup_\varphi E_2$ depends only on the homotopy class of the isomorphism $\varphi : E_1|_A \to E_2|_A$.*

Proof. A homotopy of isomorphisms is a map

$$\Phi : E_1|_A \times I \to E_2|_A \,,$$

such that for all t the map

$$\Phi_t : E_1|_A \to E_2|_A \quad \Phi_t(\xi) := \Phi(\xi, t)$$

is a bundle isomorphism. We can encode this Φ into a bundle isomorphism as follows: Let $\pi : X \times I \to X$ be the projection. Then elements $\xi \in \pi^* E_1$ are by definition 2.1.13 of the form $\xi_{x,t} = ((x,t), \xi_x) \in (\pi^* E_1)_{x,t}$. With this notation define the bundle isomorphism:

$$\Psi : \pi^*(E_1)|_{A\times I} \to \pi^*(E_2)|_{A\times I} \,, \quad \Psi(\xi_{x,t}) := ((x,t), \Phi_t(\xi_x))$$

This is a bundle isomorphism, since Φ_t is one. Now denote the inclusion

$$i_t : X \to X \times I \,, \quad i_t(x) := (x,t) \,,$$

and notice that the pullback along i_t is the same as restricting to $X \times \{t\}$.

With those definitions it is clear by the construction of the gluing, that

$$E_1 \cup_{\Phi_t} E_2 \cong i_t^* \left(\pi^*(E_1) \cup_\Psi \pi^*(E_2)\right) . \tag{1}$$

{eq:kTheo-gluingHomotopyInvariant}

The isomorphism can be seen by part (2) of Corollary 3.2.10. as follows: First notice the isomorphism $\pi^*(E_i)|_{A\times\{t\}} \cong E_i$ for $i = 1,2$ and any t. By construction of the gluing this leads to the isomorphism

$$E_1 \cup_{\Phi_t} E_2 \cong \pi^*(E_1)|_{A\times\{t\}} \cup_{(\Phi_t, \mathrm{id})} \pi^*(E_2)|_{A\times\{t\}} \,.$$

Now again by construction we notice, that gluing restricted spaces is the same as gluing the whole space and then restricting. Furthermore the map (Φ_t, id) is the restriction of Ψ to $X \times \{t\}$. Therefore:

$$\pi^*(E_1)|_{A\times\{t\}} \cup_{(\Phi_t, \mathrm{id})} \pi^*(E_2)|_{A\times\{t\}} \cong (\pi^*(E_1) \cup_\Psi \pi^*(E_2))|_{X\times\{t\}}$$

Finally, pulling back via i_t is the same as restricting $X \times i$ to $X \times \{t\}$ and identifying X with $X \times \{t\}$. This concludes the chain of arguments:

$$(\pi^*(E_1) \cup_\Psi \pi^*(E_2))|_{X\times\{t\}} \cong i_t^* \left(\pi^*(E_1) \cup_\Psi \pi^*(E_2)\right) .$$

Since the pullback is homotopy invariant we can conclude by the isomorphism (1):

$$E_1 \cup_{\Phi_0} E_2 \cong E_1 \cup_{\Phi_1} E_2 \,.$$

□

def:kTheo-homotopyClassesOfMaps

Definition 3.2.12. Let X, Y be topological spaces. By $[X, Y]$ we denote the set of homotopy classes of maps from X to Y.

lem:kTheo-gluingAndHomotopyClassesToGLn

Lemma 3.2.13. *Let $X = X_1 \cup X_2$ with X_1 and X_2 closed contractible spaces. Then there is a canonical bijection*

$$\mathrm{Vect}_n(X) \to [A, \mathrm{GL}_n(\mathbb{C})] .$$

Proof. Let $[E]$ be any vector bundle over X of rank n. Then E can be represented by a vector bundle obtained by gluing that is trivial on both X_i: Since X_i is contractible we can choose two trivializations

$$\alpha_1 : E|_{X_1} \to X_1 \times \mathbb{C}^n, \quad \text{and} \quad \alpha_2 : E|_{X_2} \to X_2 \times \mathbb{C}^n .$$

With part (2) of Corollary 3.2.10 we can conclude:

$$E \cong E|_{X_1} \cup_{\mathrm{id}_A} E|_{X_2} \cong X \times \mathbb{C}^n \cup_{\alpha_2 \circ \alpha_1^{-1}|_A} X \times \mathbb{C}^n$$

Since a map between the two trivial bundles $A \times \mathbb{C}^n \to A \times \mathbb{C}^n$ can be identified with a map $A \to \mathrm{GL}_n(\mathbb{C})$ we get an assignment from bundles over X together with the trivializations to maps $A \to \mathrm{GL}_n(\mathbb{C})$. The next step is to show that two isomorphic vector bundles give homotopic maps: Assume that we are given two vector bundles E, F over X that are isomorphic via $T : E \to F$ and without loss of generality we assume that they are given as the gluing of two trivial bundles:

$$E : X_1 \times \mathbb{C}^n \cup_\varphi X_2 \times \mathbb{C}^n , \quad \text{and} \quad F : X_1 \times \mathbb{C}^n \cup_\psi X_2 \times \mathbb{C}^n .$$

We now want to define a homotopy between psi and φ. For this we start by splitting the isomorphism T into its two parts over X_1 and X_2:

$$T|_{X_i} : X_i \times \mathbb{C}^n \to X_i \times \mathbb{C}^n .$$

These maps are given by maps

$$T_i : X_i \to \mathrm{GL}_n(\mathbb{C}) ,$$

where the gluing compatibility of the $T|_{X_i}$ yield for any $a \in A$:

$$\psi(a) T_1(a) = T_2(a) \varphi(a) , \qquad \text{equivalently} \qquad \psi(a) = T_2(a) \varphi(a) T_1^{-1}(a) .$$

By the contractibility of X_i we have for any fixed $x_i \in X_i$ a homotopy $h_t^i : X_i \to X_i$ such that $h_0^i = c_{x_i}$ is the collapsing map whilst $h_1^i = \mathrm{id}_{X_i}$. Furthermore, since $\mathrm{GL}_n(\mathbb{C})$ is connected we have a path $\gamma^i : I \to \mathrm{GL}_n(\mathbb{C})$ that connects $T_i(x_i) = \gamma^i(1)$ to $\mathbb{1}_n = \gamma^i(0)$. With this we can define the homotopies

$$\begin{aligned} T_{i,t} : X_i &\to \mathrm{GL}_n(\mathbb{C}) \\ x &\mapsto \begin{cases} T_i \circ h_{2t}^i(x) & \text{if } t \le \frac{1}{2} , \\ \gamma^i(2(t - \frac{1}{2})) & \text{if } t > \frac{1}{2} , \end{cases} \end{aligned}$$

that satisfy $T_{i,1} = T_i$ and $T_{i,0}(x) = \mathbb{1}_n$ for all x. Finally we define the homotopy $H_t : A \to \mathrm{GL}_n(\mathbb{C})$ by

$$H_t(a) := T_{2,t}(a)\varphi(a)T_{1,t}^{-1}(a)$$

to get a homotopy from $\varphi = H_0$ to $\psi = H_1$. Hence, we get a well defined map from isomorphism classes of bundles over X of rank n to the homotopy classes of maps from A to $\mathrm{GL}_n(\mathbb{C})$ that is the inverse of the gluing construction and hence is bijective. □

def:kTheo-metricsOnBundles

Definition 3.2.14 (Metrics on Bundles)**.** A **metric** on a complex bundle E is a section $h : X \to \mathrm{Herm}(E)$ such that $h(x)$ is positive definite for all x. $\mathrm{Herm}(E)$ denotes the bundle of hermitian forms (conjugate-symmetric sesquilinear) in each fiber. A bundle together with a metric is a **Hermitian bundle**.

def:kTheo-projectionOperator

Definition 3.2.15 (Projection Operator)**.** A **projection operator** $P : E \to E$ is a bundlehomomorphism in E such that $P^2 = P$.

lem:kTheo-projectionsGiveSumdecomposition

Lemma 3.2.16. *Any projection operator $P : E \to E$ determines a direct sum decomposition $E = PE \oplus (1 - P)E$.*

Proof. Observe that fibrewise $1 - P$ maps each ξ to the kernel of P, and hence fiberwise $P(E)$ and $(1-P)(E)$ are disjoint. Furthermore $\mathrm{im}(P)+\mathrm{im}(1-P) = E$, because $v = P(v)+(v - P(v))$. Therefore in each fibre we have the direct sum $E_x = P(E_x) \oplus (1 - P)(E_x)$. Now the question is, if that gives a global splitting, meaning that $P(E)$ and $(1 - P)(E)$ are continous as subbundles of E. For this to be true, we need local triviality, which is clear as long as the rank of P (and $(1 - P)$) is locally constant.In each fiber we know that $\mathrm{rank}(P_x) + \mathrm{rank}(1 - P) = \dim(E_x)$. Furthermore, $\mathrm{rank}(P_x)$ and $\mathrm{rank}(1 - P)$ are locally constant. To see this first notice that they are lower semi continuous, meaning that for y from a small enough neighbourhood of x $\mathrm{rank}(P_x) \le \mathrm{rank}(P_y)$. To see this, pick a $\mathrm{rank}(P_x)_x \times \mathrm{rank}(P_x)_x$ minor with nonzero determinant. The latter is an open condition and hence stable under small perturbation. But now we have the sum of two lower semi continuous functions that give something locally constant and hence, each summand is locally constant. Finally the above shows that $\ker(P) = (1-P)(E)$ we have the decomposition. □

lem:kTheo-metricInducedComplement

Lemma 3.2.17 (Metric Induced Complement)**.** *Suppose that F is a sub-bundle of E. Then a metric provides a canonical complementary sub-bundle.*

Proof. Let h denote the metric. For each x we have the orthonormal projection $P_x : E_x \to F_x$ inducing a map $P : E \to F$. Let $(f_1, \dots f_n)$ be an orthonormal frame of F. Then fiberwise with $\xi \in E_x$ the orthogonal projection is given by

$$P_x(\xi) = \sum_i h_x(\xi, f_i(x))f_i(x)\,.$$

And since h is continuous the whole map P is. Furthermore $P^2 = P$ and hence we get a canonical sum decomposition. □

Definition 3.2.18. We call a sequence of bundle homomorphisms

$$\cdots \longrightarrow E \longrightarrow F \longrightarrow \cdots$$

exact, if it is fiberwise exact.

lem:kTheo-splittingPrinziple

Lemma 3.2.19 (Splitting Prinziple)**.** *Every short exact sequence of bundles over the same base space splits.*

Proof. Let X be the base space and assume that the sequence is given as

$$0 \longrightarrow E' \xrightarrow{\varphi'} E \xrightarrow{\varphi''} E'' \longrightarrow 0$$

Now it is well known that any short exact sequence of vector spaces splits. We just need to ensure that this splitting is continuous. For this we give E a metric. Now since φ' is injective, $\varphi'(E')$ is a subbundle. By Lemma 3.2.17 we can split $E \cong \varphi'(E') \oplus L$. However, $\varphi''|_L$ is fiberwise (and hence as a bundle map) an isomorphism. This can be seen as φ'' being surjective and trivial on $\varphi'(E')$ hence the restriction is still surjective. The injectivity is also clear as the kernel by the homomorphism theorem of linear maps. But then we clearly have $E = E' \oplus E''$ and the sequence is isomorphic as a short exact sequence to the splitting one:

$$0 \longrightarrow E' \xrightarrow{i_1} E' \oplus E'' \xrightarrow{\pi_2} E'' \longrightarrow 0$$

□

def:kTheo-ampleSubspace

Definition 3.2.20. A subspace $V \subseteq \Gamma(E)$, where $\Gamma(E)$ is viewed as a $\mathbb{C}$-vector space is called **ample**, if

$$\varphi : X \times V \to E, \quad \varphi(x, s) := s(x)$$

is a surjection.

Corollary 3.2.21. If the subspace V is ample and $\dim(V) = \operatorname{rank}(E)$, then E is trivializable.

lem:kTheo-ampleSubspaceExists

Lemma 3.2.22. *For any vector bundle E over a compact Hausdorff space X, there exists an finite dimensional ample subspace of* $\Gamma(E)$*.*

Proof. Let $\{U_\alpha\}$ be a finite open cover of X such that $E|_{U_\alpha}$ is trivial. Let $\{s_{\alpha,i}\}$ be the corresponding local frames, and define V_α to be the finite dimensional ample subspace $V_\alpha \subseteq \Gamma(E|_{U_\alpha})$. Let ρ_α be a partition of unity subordinate to the cover $\{U_\alpha\}$. Define

$$\theta_\alpha : V_\alpha \to \Gamma(E)\,, \quad \theta_\alpha(s_\alpha)(x) = \begin{cases} s_\alpha(x) \cdot \rho_\alpha(x) & \text{if } x \in U_\alpha\,, \\ 0 & \text{else.} \end{cases}$$

Then the θ_α give rise to a homomorphism

$$\theta : \Pi_\alpha V_\alpha \to \Gamma(E)\,,$$

and the image of θ is finite dimensional and ample. The surjectivity is clear, as for any x there exists a $\rho_\alpha(x) > 0$ making the map

$$\theta_\alpha(V_\alpha) \to E_x$$

a surjection. □

cor:kTheo-epiFromTrivial

Corollary 3.2.23. *If E is any bundle over X, there exists an epimorphism $\varphi : X \times \mathbb{C}^m \to E$ for some integer m.*

Proof. Just take a basis $(s_1, \ldots, s_m)$ of an ample subspace V and define $\varphi(x, e_i) = s_i(x)$, where e_i is the i-th standard basis vector of $\mathbb{C}^m$. □

cor:kTheo-splittingTrivial

Corollary 3.2.24. *If E is a bundle over X, there exists a bundle F such that $E \oplus F$ is trivial.*

Proof. By Corollary 3.2.23 there exists an epimorphism $\varphi : X \times \mathbb{C}^m \to E$. The kernel of φ is a subbundle of $X \times \mathbb{C}^m$ and hence by Lemma 3.2.19 we have a splitting of the short exact sequence

$$0 \longrightarrow \ker(\varphi) \longrightarrow X \times \mathbb{C}^m \xrightarrow{\ \varphi\ } E \longrightarrow 0$$

Hence, the middle term becomes isomorphic to the direct sum of the outer terms and we get $X \times \mathbb{C}^m \cong E \oplus \ker(\varphi)$. □

def:kTheo-grassmanManifold

Definition 3.2.25 (Grassmann Manifold)**.** Let V be a vector space and n be any integer. The **Grassmann manifold** $\mathrm{Gr}_n(V)$ is the set of all subspaces of V with **codimenion** n. If V is given a Hermitian metric, we can topologize $\mathrm{Gr}_n(V)$ as a subspace of the set of all projection operators on V and this defines a map

$$P : \mathrm{Gr}_n(V) \to \mathrm{End}(V)\,,$$

where $\mathrm{End}(V)$ is the space of all endomorphisms of V with the operator norm topology. We give $\mathrm{Gr}_n(V)$ the subspace topology of $\mathrm{End}(V)$.

def:kTheo-classifyingMap

Definition 3.2.26. Suppose that E is a bundle over X, V is a vector space and $\varphi : X \times V \to E$ is an epimorphism. Then we can define a map $f : X \to \mathrm{Gr}_{\dim(V)}(V)$ by sending x to the kernel of $\varphi_x : V \to E_x$. We call f to be induced by φ. This map is continouos for any metric on V. To see this notice that the topology on V is metric invariant, as V is finite dimensional. Now the map f is continuous if the map for the map

$$P : \mathrm{Gr}_n(V) \to \mathrm{End}(V)$$

the composition $P \circ f$ is continuous. But $P \circ f$ is given by sending x to the projection onto $\ker(\varphi_x)$, which is continuous as φ is.

def:kTheo-classifyingBundle

Definition 3.2.27 (Classifying Bundle). Let V be a vector space and $F \subseteq \mathrm{Gr}_n(V) \times V$ be the subbundle consisting of all points (g, v) such that $v \in g$. We call F the **tautological bundle** over $\mathrm{Gr}_n(V)$. Furthermore, if $E = (\mathrm{Gr}_n(V) \times V)/F$ is the quotient bundle, we call E the **classifying bundle** over $\mathrm{Gr}_n(V)$.

Corollary 3.2.28. If E' is a bundle over X, and $\varphi : X \times V \to E'$ is an epimorphism, then φ induces a map $f : X \to \mathrm{Gr}_m(V)$ such that $E' \cong f^*E$, wehere E is the classifying bundle over $\mathrm{Gr}_m(V)$. The isomorphism follows from a fiberwise construction because $E_x \cong V \big/ \ker(\varphi_x)$. The latter however is exactly the fiber over $\ker(x)$ in the classifying bundle and hence the pullback via the φ induced map gives an isomorphism. That motivates the name "classifying bundle" for E and "classifying map" for f.

lem:kTheo-GrassmanTopologyMetricInvariant

Lemma 3.2.29. *The topology on* $\mathrm{Gr}_n(V)$ *is independent of the choice of the metric on* V.

Proof. Let h and h' be two metrics on V. Then we denote the topological space $\mathrm{Gr}_n(V)$ with the topology induced by h by $\mathrm{Gr}(V_h)$ and the one induced by h' by $\mathrm{Gr}(V_{h'})$. We need to show that the identity map $\mathrm{id} : \mathrm{Gr}(V_h) \to \mathrm{Gr}(V_{h'})$ is a homeomorphism. For this we need to show that id is continuous.The projection $\mathrm{Gr}_n(V_h) \times V_{h'} \to E$ (where E is the classifying bundle) is an epimorphism and hence induces a classifying map $f : \mathrm{Gr}_n(V_h) \to \mathrm{Gr}_n(V_{h'})$ such that $f^*E \cong E$. However that map is given by the identity on $\mathrm{Gr}_n(V)$ and hence the identity is continuous by Definition 3.2.26. □

def:kTheo-directLimitOfSets

Definition 3.2.30 (Direct Limit of Sets). Let $X_1 \to X_2 \to \cdots$ be a sequence of sets and injective maps. Then the **direct limit** $\lim_{m\to\infty} X_m$ is the union $\bigcup_{m=1}^{\infty} X_m$ modulo the equivalence relation identifying x with y if there exists an m such that $i_m(x) = i_m(y)$. We call $\lim_{m\to\infty} X_m$ the **direct limit** of the sequence $X_1 \to X_2 \to \cdots$.

def:kTheo-directLimitOfTopologicalSpaces

Definition 3.2.31 (Direct Limit of Topological Spaces). Let $X_1 \to X_2 \to \cdots$ be a sequence of topological spaces and continuous injective maps. Then the **direct limit** $\lim_{m\to\infty} X_m$ is the set theoretic direct limit with the topology given by the final topology with respect to the maps $X_m \to \lim_{m\to\infty} X_m$. I.e. a subset $U \subseteq \lim_{m\to\infty} X_m$ is open if and only if $U \cap X_m$ is open for all m.

lem:kTheo-universalPropertyOfDirectLimit

Lemma 3.2.32 (Universal Property of the Direct Limit). *Let*

$$X_1 \xrightarrow{i_1} X_2 \xrightarrow{i_2} \cdots$$

be a sequence of sets (or topological spaces). Then any family of (continuous) maps $f_m : X_m \to Y$ *that are compatible with the* i_m *(i.e.* $f_{m+1} \circ i_m = f_m$*) factors uniquely through a (continuous) map* $f : \lim_{m\to\infty} X_m \to Y$.

Proof. We start by showing the existence of f by defining $f(x) = f_m(x)$ for $x \in X_m$. This is well defined as the f_m are compatible with the i_m. Now we need to show that f is continuous. For this we need to show that $f^{-1}(U)$ is open for any open subset $U \subseteq Y$. But by definition of the topology on the direct limit, it suffices to show that $f^{-1}(U) \cap X_m$ is open for all m. This is clear as $f^{-1}(U) \cap X_m = f_m^{-1}(U)$, which is open as f_m is continuous. Finally, the uniqueness of f follows from the fact that the maps $X_m \to \lim_{m\to\infty} X_m$ are injective and their images cover $\lim_{m\to\infty} X_m$. □

Corollary 3.2.33. The inverse system given by the projections

$$\pi : \mathbb{C}^m \to \mathbb{C}^{m-1}, \quad (x_1, \dots, x_m) \mapsto (x_1, \dots, x_{m-1}),$$

induces an direct system of Grassmann manifolds with the induced continuous maps

$$i_{m-1} : \mathrm{Gr}_n(\mathbb{C}^{m-1}) \to \mathrm{Gr}_n(\mathbb{C}^m), \quad U \mapsto \pi^{-1}(U).$$

If we denote the classifying bundle over $\mathrm{Gr}_n(\mathbb{C}^m)$ by E_m, then the pullback of E_m along i_{m-1} is isomorphic to E_{m-1}.

thm:kTheory-classifyingSpace

Theorem 3.2.34. *The map*

$$\lim_{m\to\infty} [X, \mathrm{Gr}_n(\mathbb{C}^m)] \to \mathrm{Vect}_n(X),$$

induced by $f \mapsto f^* E_m$ is a bijection for all compact Hausdorff spaces X.

Proof. We shall contrstuct and inverse map. If E is a bundle over X, then by Corollary 3.2.23 there exists an epimorphism $\varphi : X \times \mathbb{C}^m \to E$ for some integer m. Let $f : X \to \mathrm{Gr}_n(\mathbb{C}^m)$ be the map induced by φ. If we manage to show that the homotopy class of f as a map to $\mathrm{Gr}_n(\mathbb{C}^{m'})$ for some large enough m' is independent of the choice of φ and m, then we get a well defined map from $\mathrm{Vect}_n(X)$ to $\lim_{m\to\infty}[X, \mathrm{Gr}_n(\mathbb{C}^m)]$ that is inverse to the classifying map.

Suppose that $\varphi_i : X \times C^{m_i} \to E$ for $i = 1, 2$ are two epimorphisms. Let $g_i : X \to \mathrm{G}_n(C^{m_i})$ be the maps induced by φ_i. We want to show that g_1 and g_2 are homotopic as maps to $\mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$. For this we define the map

$$\psi_t : X \times \mathbb{C}^{m_1} \times C^{m_2} \to E, \quad \psi_t(x, v_1, v_2) = t\varphi_1(x, v_1) + (1-t)\varphi_2(x, v_2).$$

Since the φ_i are epimorphisms, this map is an epimorphism for any t. Let

$$f_t : X \to \mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$$

be the mapes induced by ψ_t. (where we identified $\mathbb{C}^{m_1} \times \mathbb{C}^{m_2}$ with $\mathbb{C}^{m_1+m_2}$ by mapping (v_1, v_2) to (v_1, v_2)). Then

$$f_0 = j_0 \circ g_1 \quad \text{and} \quad f_1 = T \circ j_1 \circ g_2,$$

where $j_i : \mathrm{Gr}_n(\mathbb{C}^{m_i}) \to \mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$ is the map induced by the inclusion $\mathbb{C}^{m_i} \to \mathbb{C}^{m_1+m_2}$ and T is the map induced by the isomorphism $\mathbb{C}^{m_1+m_2} \to \mathbb{C}^{m_1+m_2}$ that sends (v_1, v_2) to (v_2, v_1). Since T is homotopic to the identity, we can conclude that $j_0 \circ g_1$ and $j_1 \circ g_2$ are homotopic as maps to $\mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$. Hence, the homotopy class of the map induced by an epimorphism is independent of the choice of the epimorphism and the integer m. This concludes the proof. □

Lemma 3.2.35. *For any compact X and sequence of injective maps $i_m : X_m \to X_{m+1}$ we have a canonical bijection between the set theoretic direct limit of homotopy classes of maps and the homotopy classes of maps to the topological direct limit:*

$$\lim_{m\to\infty} [X, \mathrm{Gr}_n(\mathbb{C}^m)] \cong [X, \lim_{m\to\infty} \mathrm{Gr}_n(\mathbb{C}^m)] \,.$$

Here $[X, \mathrm{Gr}_n(\mathbb{C}^m)] \to [X, \mathrm{Gr}_n(\mathbb{C}^{m-1})]$ is the map induced by $i_{m-1} : \mathrm{Gr}_n(\mathbb{C}^{m-1}) \to \mathrm{Gr}_n(\mathbb{C}^m)$.

Proof. Since X is compact, any map $f : X \to \lim_{m\to\infty} \mathrm{Gr}_n(\mathbb{C}^m)$ has image that is also compact and hence contained in some $\mathrm{Gr}_n(\mathbb{C}^m)$ for m big enough (Just cover it with open maps and reduce that cover to a finite one). Hence there exists a f_m such that $f = i \circ f_m$. where i denotes the inclusion of $\mathrm{Gr}_n(\mathbb{C}^m)$ into the direct limit. This assignement is independent of the choosen m, once we view f_m as an element of the the direct limit. This assignement is well defined on homotopy classes becuse if $f, s : X \to \lim_{m\to\infty} \mathrm{Gr}_n(\mathbb{C}^m)$ are homotopic, then there exists a homotopy $H : X \times I \to \lim_{m\to\infty} \mathrm{Gr}_n(\mathbb{C}^m)$ between them. By the same argument as before, the image of H is compact and hence contained in some $\mathrm{Gr}_n(\mathbb{C}^m)$. Hence, f_m and s_m are homotopic via H_m as maps to $\mathrm{Gr}_n(\mathbb{C}^m)$. Hence we get a well defined canonical map

$$\Phi : [X, \lim_{m\to\infty} \mathrm{Gr}_n(\mathbb{C}^m)] \to \lim_{m\to\infty} [X, \mathrm{Gr}_n(\mathbb{C}^m)] \,.$$

This map is injective, as if f_m and s_m are homotopic as maps to $\mathrm{Gr}_n(\mathbb{C}^m)$, then they are also homotopic as maps to the direct limit. Finally Φ is surjective, as any element of the direct limit is represented by some $f_m : X \to \mathrm{Gr}_n(\mathbb{C}^m)$ and hence $\Phi([i \circ f_m]) = [f_m]$. □

Definition 3.2.36. We denote $\lim_{m\to\infty} \mathrm{Gr}_n(\mathbb{C}^m)$ by $\mathrm{BU}(n)$ and call it the **classifying space** for complex vector bundles of rank n. The name origins from the fuctor B that sends a topological group G to the classifying space of vector bundles with structure group G.

Corollary 3.2.37. For any compact Hausdorff space X, there is a canonical bijection between the homotopy classes of maps from X to $\mathrm{BU}(n)$ and the isomorphism classes of vector bundles of rank n over X:

$$[X, \mathrm{BU}(n)] \cong \mathrm{Vect}_n(X) \,.$$

3.3 K-Theory as a Cohomology Theory

Definition 3.3.1. We need the following categories of topological spaces:

- **TopC** is the category of compact Hausdorff spaces.
- $\mathbf{TopC^0}$ is the category of pointed compact Hausdorff spaces.
- $\mathbf{TopC^2}$ is the category of pairs of compact Hausdorff spaces.

Here, we have the natural functors:

- $\mathbf{TopC^2} \to \mathbf{TopC^0}$ by sending a pair $(X,Y) \mapsto (X/Y,[Y])$ with obvious morphisms. If $Y = \varnothing$ this is the disjoint union of X with a point.
- $\mathbf{TopC} \to \mathbf{TopC^2}$ by sending $X \mapsto (X,\varnothing)$ again with obvious morphisms.
- $\mathbf{TopC} \to \mathbf{TopC^0}$ as the composition.

If we decent to the category where the morphisms are homotopy classes of maps, we get the categories **hTopC**, $\mathbf{hTopC^0}$ and $\mathbf{hTopC^2}$. Furthermore, the construction gives a functor from the categorie of abelien semigroups to the categorie of abelien groups.

def:kTheo-grothendiekGroupAndRing

Definition 3.3.2 (Grothendiek Group and Ring)**.** Let A be an abelien semigroup. Then the **Grothendiek group** of A is the abelian group $K(A)$ together with a homomorphism $\varphi : A \to K(A)$ such that for any other group G together with a homomorphism $\psi : A \to G$, there exists a unique homomorphism $\psi : K(A) \to G$ such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[dr, "\psi"'] & K(A) \arrow[d, dashed, "\tilde{\psi}"] \\ & G \end{tikzcd}$$

If G is a semiring, then $K(G)$ inherits a ring structure and we call $K(G)$ the **Grothendiek ring** of G.

cor:kTheo-ConstructionOfGrothendiekGroup

Corollary 3.3.3. The Grothendiek group exists for any abelian semigroup and is unique up to canonical isomorphism. If G is a semiring, then the Grothendiek ring exists and is unique up to canonical isomorphism.

Proof. Let A be an abelien semigroup and $\Delta : A \to A \times A$ be the diaglinal homomorphism that sends a to (a,a). Let $K(A) := A \times A \big/ \Delta(A)$. Then $K(A)$ is a priori an abelien semigroup, but for $a, b \in A$ we have

$$(a,b) + (b,a) = (a+b, b+a) \in \Delta(A)\,,$$

and hence $K(A)$ is an abelian group. Also we can see that

$$\overline{(a,b)} = \overline{(a,0)} - \overline{(b,0)} = \alpha_A(a) - \alpha_A(b)$$

Together with the homomorphism $\alpha_A : A \to K(A)$ that sends a to the class $(a,0)+\Delta(A)$, we get the universal property of the Grothendiek group:

Let G be another abelian group and $\psi : A \to G$ a semigroup homomorphism.From this define

$$\tilde{\psi} : K(A) \to G\,, \quad (a,b) + \Delta(A) \to \psi(a) - \psi(b)$$

The well definintion of this map can be verified by a calculation. By calculating it also follows that $\tilde{\psi}$ is a group homomorphism and that $\tilde{\psi} \circ \alpha_A = \psi$. Hence for the universal property we only need to verify uniqueness. So assume that $\theta : K(A) \to G$ is another group homomorphism. Then:

$$\theta\left(\overline{(a,b)}\right) = \underbrace{\theta(\alpha_A(a))}_{\psi(a)} - \underbrace{\theta(\alpha_A(b))}_{\psi(b)} = \tilde{\psi}\left(\overline{(a,b)}\right)$$

This concludes the construction of the grothendiek group. To see the functoriality we first need to define what $K(\beta)$ means, when $\beta : A \to B$ defines a semigroub homomorphism between to abelien semigroups. For this we use the universal property and define $K(\beta)$ to be the unique group homomorphism comming from $\alpha_B \circ \beta : A \to K(B)$. Now the uniqueness of said definition shows the equialities $K(\mathrm{id}) = \mathrm{id}$ and $K(f \circ g) = K(f) \circ K(g)$. □

Corollary 3.3.4. In the proof above we have shown that the assignement K is a functor from the category of abelian semigroups to the category of abelian groups. Alternatively, we can view K as a functor from the category of semirings to the category of rings.

Definition 3.3.5. Let X be a compact Hausdorff space. We define the **K-group** of X to be the Grothendiek ring of the semiring $\mathrm{Vect}(X)$ of isomorphism classes of vector bundles over X with the direct sum as addition and the tensor product as multiplication. We denote the K-group of X by $K(X)$. As a whole we get a contravariant functor from the category of compact Hausdorff spaces to the category of rings

$$\mathbf{TopC} \to \mathbf{Ring}\,. \tag{2}$$

Remark 3.3.6. For a continuous map $f : X \to Y$ we denote the induced map $K(f^*) : K(Y) \to K(X)$ by jusst f^*.

> **Theorem 3.3.7.** *The functor K : $\mathbf{TopC} \to \mathbf{Ring}$ factors over the homotopy category* **hTopC** *of compact Hausdorff spaces together with homotopy classes of maps.*

Proof. This directly follows from Theorem 3.2.4. □

def:kTheo-reducedKTheory

Definition 3.3.8 (Reduced K-Theory)**.** Let X be a compact Hausdorff space and $x_0 \in X$ be a base point. We define the **reduced K-group** $\tilde{K}(X)$ of X to be the kernel of the map $K(X) \to K(\{x_0\})$ induced by the inclusion $\{x_0\} \to X$.

cor:kTheo-splittingReducedGroupNatural

Corollary 3.3.9. If $c : X \to x_0$ is the collapsing map sending each point to the basepoint, then we get a natural splitting

$$K(X) \cong \tilde{K}(X) \oplus K(x_0)\,.$$

Naturallity here means, that for any homomorphism of pointes spaces $f : (X, x_0) \to (Y, y_0)$ we have a commuting diagramm:

$$\begin{tikzcd} K(X) \arrow[r] \arrow[d, "f^*"] & \tilde{K}(X) \oplus K(x_0) \arrow[d, "\tilde{K}(f)\oplus(\{x_0\}\to\{y_0\})^*"] \\ K(Y) \arrow[r] & \tilde{K}(Y) \oplus K(y_0) \end{tikzcd}$$

Proof. We have the exact sequence of Rings:

$$0 \longrightarrow \tilde{K}(X) \xrightarrow{inclusion} K(X) \xrightarrow{i_X^*} K(x_0) \longrightarrow 0$$

where $c_X^* : K(x_0) \to K(X)$ is a right inverse of i_X^*, and hence the sequence splits and $K(X) \cong \tilde{K}(X) \oplus K(x_0)$.The isomorphism is explicitly given by

$$K(X) \to \tilde{K}(X) \oplus K(x_0)\,, \quad \xi \mapsto (\xi - c_X^* \circ i_X^*(\xi), i_X^*(\xi))$$

and its inverse maps $(a, b) \mapsto a + c_X^*(b)$. The splitting is natural with respect to maps in $\mathbf{TopC^0}$, because for a map of pointed spaces $f : X \to Y$ we get commutativity:

diag:kTheo-NaturalSplitting

$$\begin{tikzcd} 0 \arrow[r] & \tilde{K}(X) \arrow[r, hook] & K(X) \arrow[r, "i_X^*"'] & K(x_0) \arrow[r] \arrow[l, bend right, "c_X^*"'] & 0 \\ 0 \arrow[r] & \tilde{K}(Y) \arrow[r, hook] \arrow[u, "f^*|_{\tilde{K}(Y)}"] & K(Y) \arrow[r, "i_Y^*"] \arrow[u, "f^*"] & K(y_0) \arrow[r] \arrow[u, "(f|_{x_0})^*"'] \arrow[l, bend left, "c_Y^*"] & 0 \end{tikzcd}$$

The well definition of $f^*|_{\tilde{K}(Y)}$ follows immidiatly from the commutativity. The commutativity itself follows from K beeing a contravariant functor and all topological maps commute. Finally, define $\tilde{K}(f) := f^*|_{\tilde{K}(Y)}$ and we are done, since the naturality of the splittings, i.e. commutativity of the diagram in the corollary, then follows from commutativity of each square separately: the left square controls the $\tilde{K}$-summand and the right square (together with the sections c_X^*, c_Y^*) controls the $K(x_0)$-summand. □

Remark 3.3.10. For a map $f : (X, x_0) \to (Y, y_0)$ we denote $\tilde{K}(f)$ and $K(f)$ by f^*. The context will make clear if we work in the reduced theory or not.

Corollary 3.3.11. $\tilde{K}$ is a functor from $\mathbf{TopC^0}$ to $\mathbf{Ring}$.

Proof. The proof follows imidiatly from the naturality of the splitting in Corollary 3.3.9. □

cor:kTheo-unpointedReducedIsNonReduced

Corollary 3.3.12. Let X be a compact hausdorff space, and let

$$F : \mathbf{TopC} \to \mathbf{TopC^0}$$

be the functor that that sends X to $(X, [\varnothing])$ with obvious homomorphisms. Then there is a natural isomorphism between the functors $\tilde{K} \circ F$ and K.

Proof. This follows from the fact that disjoint uninons give rise to direct sums of semi rings: $\mathrm{Vect}(X \sqcup \{\varnothing\}) \cong \mathrm{Vect}(X) \oplus \mathrm{Vect}(\{\varnothing\})$ and the isomorphism is given by gluing the bundles and hence natural. Furthermore, the Grothendiek ring of a direct sum gives the direkt sum of rings (again in a natural manner). (This can be verified using the universal property). Hence we get the commutative diagram

$$\begin{tikzcd}
0 \arrow[r] \arrow[d, equal] & \tilde{K}(F(X)) \arrow[r, "inclusion"] & K(F(X)) \arrow[r, "i^*"] \arrow[d, "\Phi"] & K([\varnothing]) \arrow[r] \arrow[d, equal] & 0 \arrow[d, equal] \\
0 \arrow[r] & K(X) \arrow[r, "i_1"] & K(X) \oplus K([\varnothing]) \arrow[r, "\pi_2"] & K([\varnothing]) \arrow[r] & 0
\end{tikzcd}$$

But then we can define the map $\Phi|_{\tilde{K}(F(X))}$ and diagram yoga shows that this is an isomorphism. Furthermore, since Φ is natural, the new map is also natural and defines a netural transformation between the two functors. $\square$

Definition 3.3.13. Let $F : \mathbf{TopC^2} \to \mathbf{TopC^0}$ be the functor that collapses the second space. Then the composition $\tilde{K} \circ F$ defines a contravariant functor.

Definition 3.3.14. Let (X, x_0) and (Y, y_0) be two pointed spaces. Then we define the **pointed sum** as:

$$X \vee Y := (X \sqcup Y) \big/ (x_0 \sim y_0)\,.$$

Whith this, we can define the **smash product** as:

$$X \wedge Y := (X \times Y) \big/ (X \vee Y)$$

Definition 3.3.15. For the sake of intuition, I want to give a bit of insight into tensor products without beeing to technical and opening the box of pandora of monoidial categories. We define a tensor product in a caregory simply as the space over which bilinear maps factor uniquely. To capture bilinearity in such a categorie wie define the following. Let $\pi_i : X_0 \times X_1 \to X_i$ be the canonical projection for $i = 0, 1$. Any resonalble product should come with such surjective maps. Since those are surjective, the π_i omitt a family of right inverses $r_i : X_i \to X_0 \times X_1$. We call such a right inverse "product preserving" if the compositions $\pi_2 \circ r_1$ and $\pi_1 \circ r_2$ are constant. Hence the right invers maps into just one fiber. Now bilinearity in this general setting states, that for any product preserving right invers of the projections r_i, the composition $f \circ r_i$ is structure preserving.

Remark 3.3.16 (What is the Smash Product)**.** We can think of the smash product as the tensor product in the pointed topological spaces as follows. Let $f : (X, x_0) \times (Y, y_0) \to (Z, z_0)$ be bilinear. For any $y \in Y$ we have the product preserving right inverse of π_1 given by

$$r_1^y : (X, x_0) \to (X, x_0) \times (Y, y_0), \quad x \mapsto (x, y) .$$

Analogously we define r_2^x for any $x \in X$ as

$$r_2^x : (Y, y_0) \to (X, x_0) \times (Y, y_0), \quad y \mapsto (x, y) .$$

Now for any $x \in X$ we have $f(x, y_0) = f \circ i_Y^x(y_0) = z_0$ as the composition is said to be basepoint preserving. Similar we see that $f(x_0, y) = z_0$ for any $y \in Y$. Hence f factors over the quotient space

$$X \times Y \big/ (X \times \{y_0\} \cup (Y \times \{x_0\})) ,$$

where the nominator is the canonically embedded $X \vee Y$ into the produc. Furtermore by the universal property of quotiens spaces, this factorization is unique.

Example 3.3.17. Let S^n be the n-dimensional spere given by all points of norm one in $\mathbb{R}^n{+}1$. Then a general fact is that $S^n \cong I^n \big/ \partial I^n$ and hence for all $n, m > 0$ we have $S^n \wedge S^m \cong S^{n+m}$. Later we wil make this homeomorphism canonical. ref

Definition 3.3.18. Let (X, x_0) be a pointed topological space. We define the **reduced suspension** of X to be

$$\Sigma X := S^1 \wedge X .$$

Furtermore, we define inductively $\Sigma^n X := \Sigma(\Sigma^{n-1} X) \cong S^n \wedge X$ and using the principle of permanence we extend this definition to $\Sigma^0 X := S^0 \wedge X \cong X$. The last homeomorphism is canonical.

Corollary 3.3.19.

def:kTheo-lowerKGroub

Definition 3.3.20 (Lower K-groups)**.** For $n \geq 0$ we define:

$$\begin{array}{lcll} \tilde{K}^{-n}(X, x_0) & := & \tilde{K}(\Sigma^n \wedge X) & \text{für} \quad (X, x_0) \in Ob(\mathbf{TopC^0}), \\ K^{-n}(X, Y) & := & \tilde{K}^{-n}(X/Y, [Y]) & \text{für} \quad (X, Y) \in Ob(\mathbf{TopC^2}) \\ K^{-n}(X) & := & K^{-n}(X, \varnothing) & \text{für} \quad X \in Ob(\mathbf{TopC}) . \end{array}$$

All those definitions give contravariant and homotopically invariant functors into the category of rings.

Corollary 3.3.21. Let X be a topological space. Then we have a natural isomorphism between the functors $K^0(X)$ and $K(X)$.

Proof. When disentangeling the definitions, we see that

$$K^0(X) = \tilde{K}(\Sigma^0(X, [\varnothing])) \cong \tilde{K}(X, [\varnothing]) \cong K(X)$$

where the first isomorphism comes from Σ^0 beeing canonically isomporphic to the identity and the second isomorphism is given in Corollary 3.3.12. $\square$

3.4 Homotopy-Theoretic definition of K

Definition 3.4.1. for any natural number $n \in \mathbb{N}$ we define $\underline{n}$ to be the n-dimensional trivial bundle over a base space X. With this we define the inclusions

$$\mathrm{Vect}_n(X) \to \mathrm{Vect}_{n+1}(X)\,, \quad \xi \mapsto \xi \oplus [\underline{1}],$$

and this gives rise to the direct limit $\lim_n \mathrm{Vect}_n(X)$.

Definition 3.4.2 (Stabely Equivalence)**.** Two vector bundles E, F over the same base space are called **stably equivalent** if there exists an n such that E and F become isomorphic after adding the trivial bundle:

$$E \oplus \underline{n} \cong F \oplus \underline{n}\,.$$

Theorem 3.4.3. *For a compact connected space X there is a natural isomorphism*

$$K(X) \cong \mathbb{Z} \times \lim_n \mathrm{Vect}_n(X)\,.$$

Proof. We already saw in Corollary 3.3.3 that any element of $K(X)$ is given by the difference of two isomorphism classes of vector bundles (included via the natural inclusion). Now let K be a arbitrary element of $K(X)$ that is represented by the difference of the isomorphism classes of the bundles E and F and let G be a bundle such that $E \oplus G$ is trivial. The latter exists by Corollary 3.2.24. Hence, we have:

$$K = [E] - [F] = ([E] + [G]) - ([F] + [G]) = ([E] + [G]) - [\underline{n}]\,,$$

for some natural number n. □

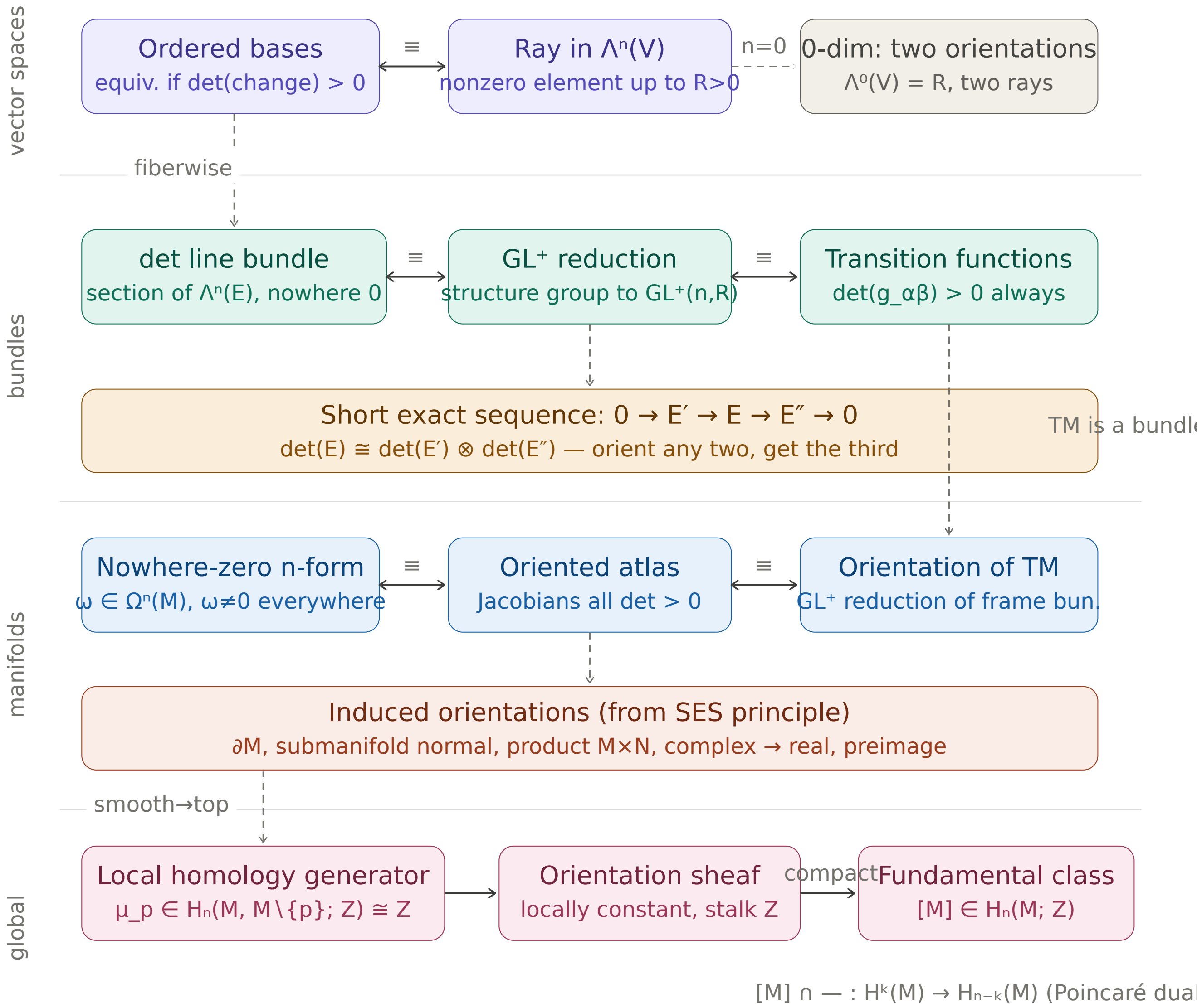
Definition 3.4.4. We define the **unitary group** U to be the direct limit of the unitary goupf of rank n under the standart inclusions

$$U(n) \to U(n+1)\,, \quad A \mapsto A \oplus \mathbb{1}_1 = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}.$$

4 Orientations

4.1 Mathematical Goals

Theorem 4.1.1 (The Theorems and Mathematical Structures).
- *Define Orientations of Manifolds*
- *How to induce Orientations*



The approach to this chapter is ...

5 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book banyaga2004lectures [BH04]. In Morse theory, we will always deal with a smooth function $f : M \to \mathbb{R}$ from a smooth Riemannian manifold (M, g) that is also compact. We will denote its dimension by $m = \dim(M)$.

Definition 5.0.1 (Critical Points)**.** A point $p \in M$ at which the smooth map $\mathrm{d}f_p = 0$ is called **critical**. By $\mathrm{Crit}(f)$ we denote the **set of critical points**.

Definition 5.0.2 (The Hessian)**.** Let $f : M \to \mathbb{R}$ be a smooth function. The **Hessian** of f at $p \in \mathrm{Crit}(f)$ is a symmetric bilinear form

$$\mathrm{Hess}_p(f) : T_pM \times T_pM \to \mathbb{R}$$

that is defined as follows. Let $\xi_p, \zeta_p \in T_pM$ be two tangent vectors. Now choose two vector fields Ξ and Z in a neighbourhood of p such that $\Xi(p) = \xi_p$ and $Z(p) = \zeta_p$. We then define the Hessian to be:

$$\mathrm{Hess}_p(f)(\xi_p, \zeta_p) = \Xi\left(Z(f)\right)(p)$$

See 2.1.33 for the notation of applying a function to a vector field. In corollary 5.0.5 we will give a local description telling us that this definition is well defined, meaning independent of the Extensions Ξ and Z.

Definition 5.0.3. A smooth function $f : M \to \mathbb{R}$ is called **Morse**, if for any critical point, the Hessian is a symmetric bilinear and non-degenerate form.

For such a form there exists the following theorem:

Theorem 5.0.4 (Sylvester's Law of Inertia)**.** *Let $A \in \mathrm{Mat}(n, \mathbb{R})$ be symmetric and let $T, T' \in \mathrm{GL}(n, \mathbb{R})$ and $k, k', l, l' \in \mathbb{N}$ such that:*

$$T^\top \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^\top \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

then $k = k'$, $l = l'$ and $\mathrm{rank}(A) = k + l$.

Proof. Since T, T' are invertible we have that

$$k + l = \mathrm{rank}(T^\top \circ A \circ T) = \mathrm{rank}(A) = \mathrm{rank}(T'^\top \circ A \circ T') = k' + l' \,. \tag{3}$$

Hence it suffices to show that $k = k'$. Which we will do by proving the claim:

$$k = \max\left\{\dim(U) | U \subseteq \mathbb{R}^n \text{subspace such that } x^\top A x > 0 \ \forall x \in U \smallsetminus \{0\}\right\}$$

So we start by showing "≤" : Denote the first k columns of T with $x_1, ..., x_k$. They form a basis of the subspace $\mathbb{R}^k \subseteq \mathbb{R}^n$ and with $0 \neq x = \sum_{i=1}^k \lambda_i x_i$ we have that by bilinearity

$$x^\top Ax = \sum_{i=1}^k \lambda_i x_i^\top Ax = \sum_{i,j=1}^k \lambda_i \lambda_j x_i^\top A x_j = \sum_{i=1}^k (\lambda_i)^2 \geq 0\,. \tag{4}$$

This concludes the first inequality.

Now let U be any k-dimensional subspace such that for all non zero $x \in U$, $x^\top Ax > 0$. By a calculation analogous to the one above we have that for any $x \in W := \operatorname{span}(x_{k+1}, \ldots, x_n)$ the number $x^\top Ax$ is less than or equal to zero. Hence, $W \cap U = \{0\}$ and with this we conclude:

$$\dim(U) = \dim(U + W) - \dim(W) + \dim(U \cap W) \leq (k + (n - k)) - (n - k) + 0 = k\,. \tag{5}$$

This completes the proof. □

cor:morse-localHessian

Corollary 5.0.5. Let $\varphi : U \to V$ be a chart around a critical point $p \in M$. Then in said chart the matrix corresponding to the Hessian has the form:

$$M_p(f) = \left(\frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j} \right)_{i,j}$$

Proof. This follows by direct calculation of $\operatorname{Hess}_p(f)\,(\partial_i(p), \partial_j(p))$. Denote $x := \varphi(p)$, then:

$$\begin{aligned}
\left(\partial_i(p)\,(\partial_j(f))_p\right) &= \left(\partial_i(p) \left(p \mapsto \left.\frac{\partial f \circ \varphi^{-1}}{\partial x_j}\right|_{\varphi(p)} \right)_p \right) \\
&= \left.\frac{\partial}{\partial x_i} \left(y \mapsto \left.\frac{\partial f \circ \varphi^{-1}}{\partial x_j}\right|_y \right)\right|_x \\
&= \frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}\,.
\end{aligned}$$

□

Remark 5.0.6. If we have two different charts, then the matrix $M_p(f)$ of the Hessian transforms from one basis to the other via conjugation of the differential of the transition function $\varphi_{12} := \varphi_1 \circ \varphi_2^{-1}$:

$$M_p^1(f) = (\mathrm{d}\varphi_{12})^\top \cdot M_p^2(f) \cdot \mathrm{d}\varphi_{12}$$

Hence we can define the following:

Definition 5.0.7. By Sylvester's Law of inertia, the number l that denotes the dimension of the maximal subspace on which a matrix is negative definite, is an invariant under conjugation. We call that number the index of said matrix. Now in the setting of Morse theory we assign to each critical point p the index of $M_p(f)$ and call that the **(Morse-)index** of p, denoted by λ_p. If omitting the function would cause ambiguity, we write λ_p^f. We denote the set of all critical points of index k by $\operatorname{Crit}_k(f)$.

thm:morse-morseLemma

Theorem 5.0.8 (Morse Lemma)**.** *Let p be a critical point of index k of the Morse function $f : M \to \mathbb{R}$. There exists a centered chart $\varphi : U \to \mathbb{R}^m$ around p such that*

$$(f \circ \varphi^{-1})(x_1, \dots, x_m) = f(p) + \sum_{i=1}^{k} -x_i^2 + \sum_{i=k+1}^{m} x_i^2 .$$

Proof. We refer to Lemma 3.11 in banyaga2004lectures [BH04] for a proof. ◻

6 Title

6.1 Mathematical Goals

Theorem 6.1.1 (The Theorems and Mathematical Structures).

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The approach to this chapter is ...

7 Title

7.1 Mathematical Goals

Theorem 7.1.1 (The Theorems and Mathematical Structures).

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